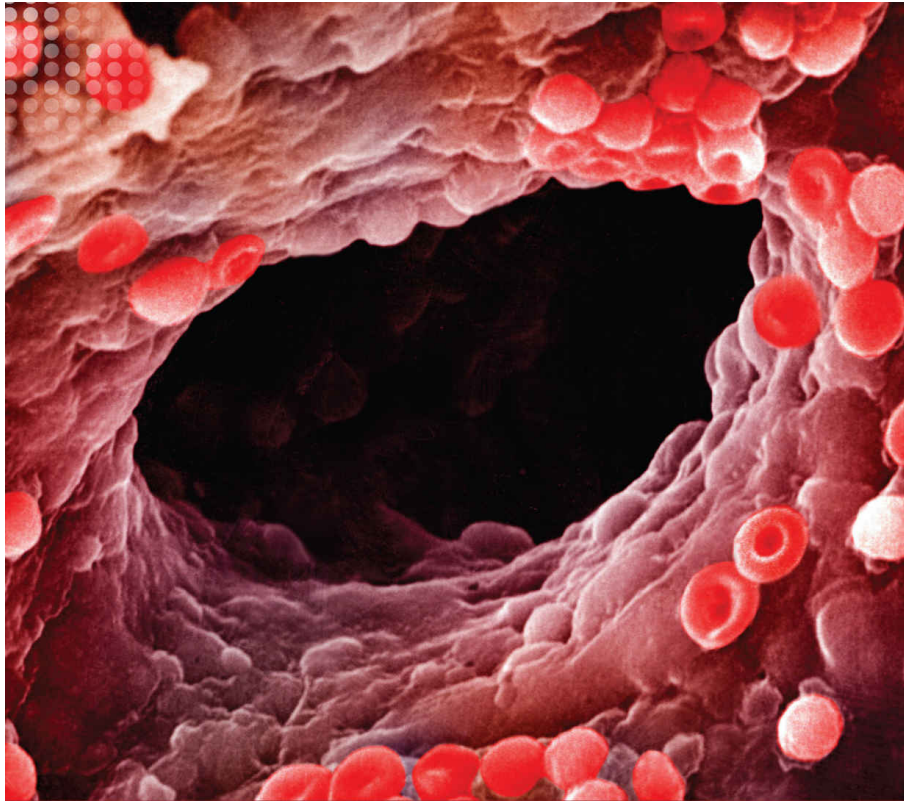


PART II

Cardiovascular and Respiratory Function

In [part I](#) of this book, we learned how skeletal muscle contracts in response to neural signaling and how the body produces energy through metabolism to fuel its movement. We also examined hormonal control of metabolism, of body fluid and electrolyte balance, and of caloric intake. Finally, we looked at how energy expenditure is measured and the causes of fatigue, soreness, and cramps. [Part II](#) focuses on how the cardiovascular and respiratory systems provide oxygen and fuel to the active muscles, how they rid the body of carbon dioxide and metabolic wastes, and how these systems respond in an integrated fashion during exercise. In [chapter 6](#), The Cardiovascular System and Its Control, we look at the structure and function of the cardiovascular system: the heart, blood vessels, and blood. In [chapter 7](#), The Respiratory System and Its Regulation, we examine the mechanics and regulation of breathing, the process of gas exchange in the lungs and at the muscles, and how oxygen and carbon dioxide are transported to muscles and other tissues in the blood. We also see how this system regulates the body's pH within a very narrow range. Finally, in [chapter 8](#), Cardiorespiratory Responses to Acute Exercise, we concentrate on the cardiovascular and respiratory changes that occur during an acute bout of exercise.





6

The Cardiovascular System and Its Control

In this chapter and in the web study guide

The Heart

Blood Flow Through the Heart

The Myocardium

The Cardiac Conduction System

The Cardiac Cycle

Determinants of Cardiac Output

ANIMATION FOR FIGURE 6.1 illustrates the course of blood flow through the human heart.

ACTIVITY 6.1 *Anatomy of the Heart* reviews the names and locations of the structures of the heart.

ACTIVITY 6.2 *Functioning of the Heart* describes blood flow through the heart and differentiates the heart's functions.

AUDIO FOR FIGURE 6.2 describes the mechanism of contraction in a cardiac muscle fiber.

ACTIVITY 6.3 *Cardiac Conduction* explores the function of each of the components of the heart's conduction system.

AUDIO FOR FIGURE 6.6 describes the contributions of the sympathetic and parasympathetic nervous systems to the rise in heart rate during exercise.

AUDIO FOR FIGURE 6.8 describes the Wiggers diagram.

VIDEO 6.1 presents Ben Levine on torsional contraction of the heart muscle and its role in ventricular filling.

Vascular System

Blood Pressure

General Hemodynamics

Distribution of Blood

ACTIVITY 6.4 *Control of the Vascular System* explains the role that parts of the vascular system play to guarantee an adequate blood supply where it is most needed.

AUDIO FOR FIGURE 6.11 describes the distribution of cardiac output at rest and during maximal exercise.

AUDIO FOR FIGURE 6.12 describes intrinsic control of blood flow.

ANIMATION FOR FIGURE 6.15 shows the action of the muscle pump.

Blood

Blood Volume and Composition

Red Blood Cells

Blood Viscosity

In Closing

R

od Williams was a 17-year-old high school junior, an offensive lineman on the football team. On September 22, 2015, Williams collapsed on the football field with no heartbeat or respirations. Despite successful CPR and hospitalization, he died 2 weeks later. Like many tragic, sudden cardiac arrest deaths in young athletes, an autopsy revealed that Williams had a preexisting heart condition that went undetected. In fact, sudden cardiac arrest is the leading cause of death in high school athletes, resulting from underlying heart anomalies that were exposed only during intense physical activity. The most common of these is hypertrophic cardiomyopathy, a genetic disease of the heart muscle, but many other causes exist, such as long QT syndrome, an electrical abnormality. Most young people living with these problems have no symptoms. Sadly, the initial manifestation is sudden cardiac arrest. While routine intensive screening seems like the logical answer, the type of advanced screening that is necessary has significant financial constraints, since well over 1 million high school football players are competing in the United States at any one time. Further, because of the low incidence of these abnormalities in the young, healthy population, the incidences of false positives (the test shows a problem where there is none) and false negatives (the test is normal but the problem actually exists) limit the tests' predictive value. Hopefully, more accurate and cost-effective screening tools are on the horizon.

The cardiovascular system serves a number of important functions in the body and supports every other physiological system. Major cardiovascular functions can be grouped into six categories:

- Delivery of oxygen and energy substrates
- Removal of carbon dioxide and other metabolic waste products
- Transport of hormones and other molecules
- Support of thermoregulation and control of body fluid balance
- Maintenance of acid–base balance to help control the body's pH
- Regulation of immune function

Although this is just an abbreviated list of roles, the cardiovascular functions listed here are important for understanding the physiological basis of exercise and sport. Obviously these roles

change and become even more critical with the challenges imposed by exercise.

All physiological functions and virtually every cell in the body depend in some way on the cardiovascular system. Any system of circulation requires three components:

- A pump (the heart)
- A system of channels or tubes (the blood vessels)
- A fluid medium (the blood)

In order to keep blood circulating, the heart must generate sufficient pressure to drive blood through the continuous network of blood vessels in a closed-loop system. Thus, the primary goal of the cardiovascular system is to ensure that there is adequate blood flow throughout the circulation to meet the metabolic demands of the tissues. We look first at the heart.

The Heart

About the size of a fist and located in the center of the thoracic cavity, the heart is the primary pump that circulates blood through the entire cardiovascular system. As shown in [figure 6.1](#), the heart has two atria that act as receiving chambers and two ventricles that serve as the pumping chambers. It is enclosed in a tough membranous sac called the **pericardium**. The thin cavity between the pericardium and the heart is filled with pericardial fluid, which reduces friction between the sac and the beating heart.

Blood Flow Through the Heart

The heart is sometimes considered to be two separate pumps, with the right side of the heart pumping deoxygenated blood to the lungs through the pulmonary circulation and the left side of the heart pumping oxygenated blood to all other tissues in the body through the systemic circulation. Blood that has circulated through the body, delivering oxygen and nutrients and picking up waste products, returns to the heart through the great veins—the superior vena cava and inferior vena cava—to the right atrium. This chamber receives all the deoxygenated blood from the systemic circulation.

From the right atrium, blood passes through the tricuspid valve into the right ventricle. This chamber pumps the blood through the pulmonary valve into the pulmonary artery, which carries the blood to the lungs. Thus, the right side of the heart is known as the pulmonary side, sending the blood that has circulated throughout the body into the lungs for reoxygenation.

After blood is oxygenated in the lungs, it is transported back to the heart through the pulmonary veins. All freshly oxygenated blood is received from the pulmonary veins by the left atrium. From the left atrium, the blood passes through the mitral valve into the left ventricle. Blood leaves the left ventricle by passing through the aortic valve into the aorta and is distributed to the systemic circulation. The left side of the heart is known as the systemic side. It receives the oxygenated blood from the lungs and then sends it out to supply all other body tissues.

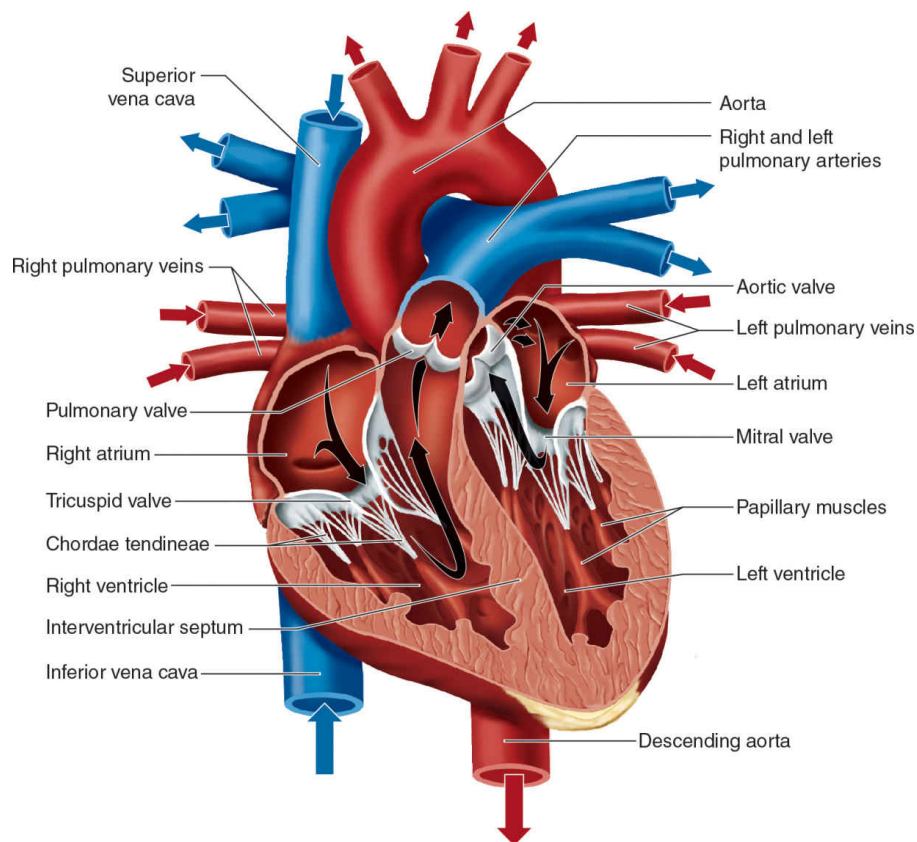


FIGURE 6.1 An anterior (as if the person is facing you) cross-sectional view of the human heart.



The four heart valves prevent backflow of blood, ensuring one-way flow through the heart. These valves maximize the amount of blood pumped out of the heart during contraction. A **heart murmur** is a condition in which abnormal sounds are detected with the aid of a stethoscope. This abnormal sound can indicate the turbulent flow of blood through a narrowed valve (stenosis) or retrograde flow back toward the atria through a leaky valve (prolapse). When valves leak as a result of disease, this condition can require surgical replacement of the valve. With *mitral valve prolapse*, the mitral valve allows some blood to flow back into the left atrium during ventricular contraction. This disorder, relatively common in adults (6%-17% of the population), usually has little clinical significance unless there is significant backflow.

Mild heart murmurs are fairly common in growing children and adolescents. Likewise, most murmurs heard in athletes are benign, affecting neither the heart's pumping nor the athlete's performance. Only when there is a functional consequence, such as light-headedness or dizziness, are murmurs a cause for immediate concern.

The Myocardium

Cardiac or myocardial muscle is collectively called the **myocardium**. Myocardial thickness at various locations in the heart varies according to the amount of stress regularly placed on the myocardium. The left ventricle is the most powerful pump because it must generate sufficient pressure to pump blood through the entire body. When a person is sitting or standing, the left ventricle must

contract with enough force to overcome the effect of gravity, which tends to pool blood in the lower extremities.

Because the left ventricle must generate considerable force to pump blood to the systemic circulation, it has the thickest muscular wall compared with the other heart chambers. This hypertrophy is the result of the pressure placed on the left ventricle at rest or under normal conditions of moderate activity. With more vigorous exercise—particularly intense aerobic activity, during which the working muscles' need for blood increases considerably—the demand on the left ventricle to deliver blood to exercising muscles is much higher. In response to both intense aerobic and resistance training, the left ventricle will hypertrophy. In contrast to this positive adaptation that occurs as a result of exercise training, cardiac muscle also hypertrophies as a result of several diseases, such as high blood pressure or valvular heart disease. In response to either training or disease, over time the left ventricle adapts by increasing its size and pumping capacity, similar to the way skeletal muscle adapts to physical training. However, the mechanisms for adaptation and cardiac performance with disease are different from those observed with aerobic training.

Although striated in appearance, the myocardium differs from skeletal muscle in several important ways. First, because the myocardium has to contract as if it were a single unit, individual cardiac muscle fibers are anatomically interconnected end to end by dark-staining regions called **intercalated disks**. These disks have desmosomes, which are structures that anchor the individual cells together so that they do not pull apart during contraction, and gap junctions, which allow rapid transmission of the action potentials that signal the heart to contract as one unit. Secondly, the myocardial fibers are rather homogeneous in contrast to the mosaic of fiber types in skeletal muscle. The myocardium contains only one fiber type, similar to type I fibers in skeletal muscle in that it is highly oxidative, has a high capillary density, and has a large number of mitochondria.

In addition to these differences, the mechanism of muscle contraction also differs between skeletal and cardiac muscle. Cardiac muscle contraction occurs by calcium-induced calcium

release (figure 6.2). The action potential spreads rapidly along the myocardial sarcolemma from cell to cell via gap junctions and also to the inside of the cell through the T-tubules. Upon stimulation, calcium enters the cell by the dihydropyridine receptor in the T-tubules. Unlike what happens in skeletal muscle, the amount of calcium that enters the cell is not sufficient to directly cause the cardiac muscle to contract, but it serves as a trigger to another type of receptor, called the ryanodine receptor, to release calcium from the sarcoplasmic reticulum. Figure 6.3 summarizes some of the similarities and differences between cardiac and skeletal muscle.

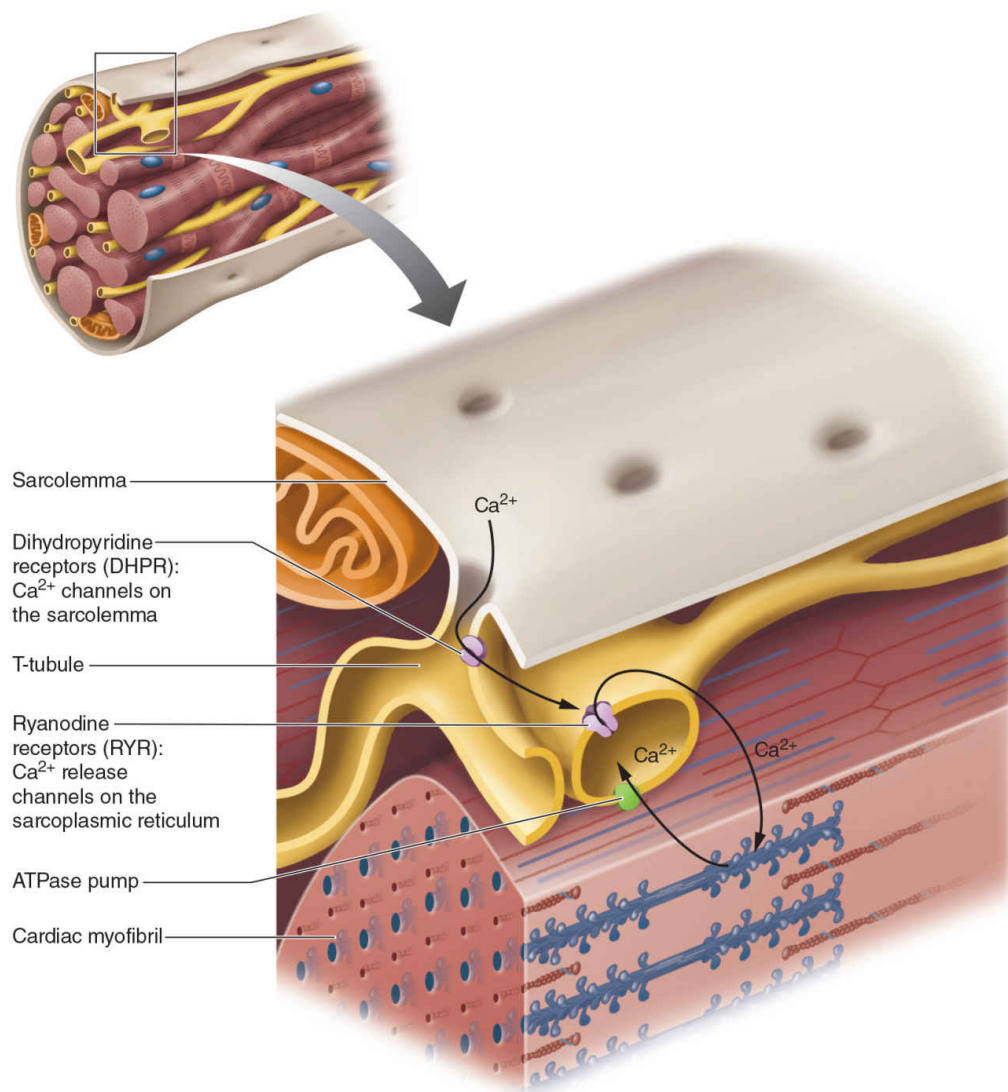


FIGURE 6.2 The mechanism of contraction in a cardiac muscle fiber, termed calcium-induced calcium release.



The myocardium, just like skeletal muscle, must have a blood supply to deliver oxygen and nutrients and remove waste products. Although blood courses through each chamber of the heart, little nourishment comes from the blood within the chambers. The primary blood supply to the heart is provided by the coronary arteries, which arise from the base of the aorta and encircle the outside of the myocardium (figure 6.4). The right coronary artery supplies the right side of the heart, dividing into two primary branches, the marginal artery and the posterior interventricular artery. The left coronary artery, also referred to as the left main coronary artery, also divides into two major branches, the circumflex artery and the anterior descending artery. The posterior interventricular artery and the anterior descending artery merge, or anastomose, in the lower posterior area of the heart, as does the circumflex. Blood flow increases through the coronary arteries when the heart is between contractions (during diastole).

The mechanism of blood flow to and through the coronary arteries is quite different from that of blood flow to the rest of the body. During contraction, when blood is forced out of the left ventricle under high pressure, the aortic valve is forced open. When this valve is open, its flaps block the entrances to the coronary arteries. As the pressure in the aorta decreases, the aortic valve closes, and blood can then enter the coronary arteries. This design ensures that the coronary arteries are spared the very high blood pressure created by contraction of the left ventricle, thus protecting these critical vessels from damage.

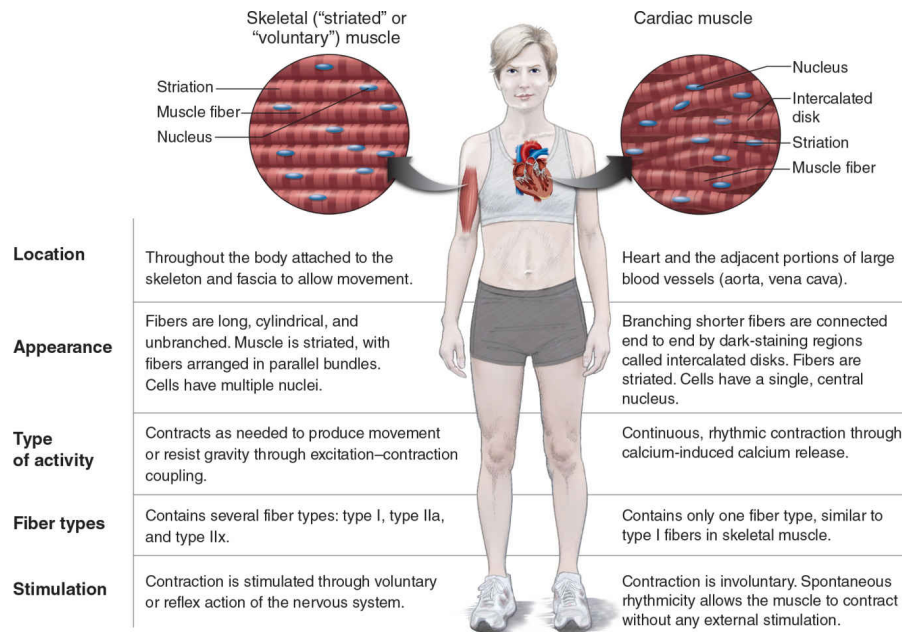


FIGURE 6.3 Functional and structural characteristics of skeletal and cardiac muscle.

The coronary arteries are, however, very susceptible to **atherosclerosis**, or narrowing by the accumulation of plaque and inflammation, leading to coronary artery disease. This disease is discussed in greater detail in [chapter 21](#). Anomalies—vessel shortenings, blockages, or flow misdirections—sometimes occur in the coronary arteries, and such congenital abnormalities are a common cause of sudden death in athletes.

In addition to its unique anatomical structure, the ability of the myocardium to contract as a single unit also depends on initiation and propagation of an electrical signal through the heart, the cardiac conduction system.

The Cardiac Conduction System

Myocardial cells are unique in that they have the ability to spontaneously depolarize and directionally conduct that electrical signal throughout the heart. The rate of depolarization is set by depolarization of a unique type of myocardial cells located in the upper right atrium and also determined by extrinsic influences, including the autonomic nervous system and circulating hormones. The following sections describe the intrinsic and extrinsic mechanisms that combine to determine heart rate and rhythm at rest and during exercise.

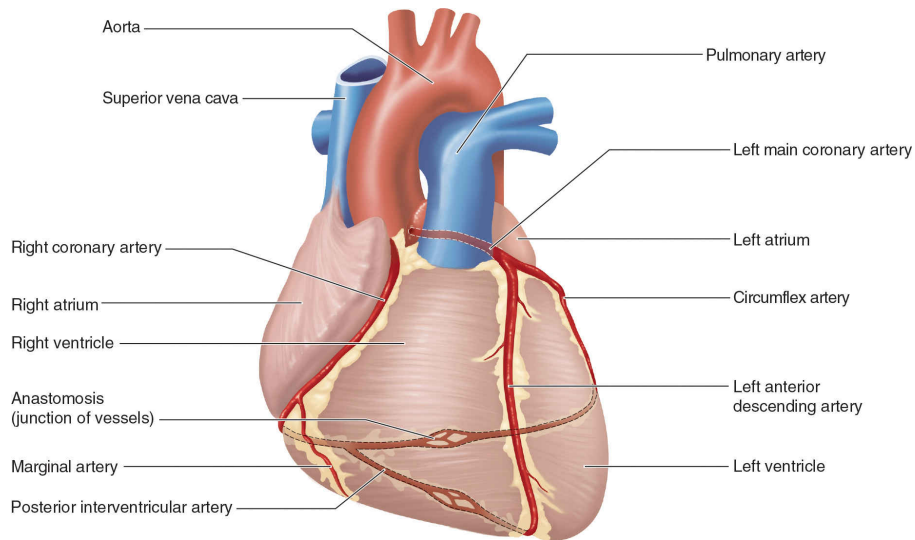


FIGURE 6.4 The coronary circulation, illustrating the right and left coronary arteries and their major branches.

Intrinsic Control of Electrical Activity

Cardiac muscle has the unique ability to generate its own electrical signal, called spontaneous rhythmicity, which allows it to contract without any external stimulation. The contraction is rhythmical, in part because of the anatomical coupling of the myocardial cells through gap junctions. Without neural or hormonal stimulation, the intrinsic heart rate averages ~100 beats (contractions) per minute. This resting heart rate of about 100 beats/min can be observed in patients who have undergone cardiac transplant surgery, because their transplanted hearts lack autonomic innervation.

Even though all myocardial fibers have inherent rhythmicity, the heart has a series of specialized myocardial cells that function to coordinate the heart's excitation and contraction and maximize the efficient pumping of blood. These are specialized cardiac muscle fibers, and not nerve tissue, even though they function to generate and transmit a signal. [Figure 6.5](#) illustrates the four main components of the cardiac conduction system:

- Sinoatrial (SA) node
- Atrioventricular (AV) node
- AV bundle (bundle of His)
- Purkinje fibers

The impulse for a normal heart contraction is initiated in the **sinoatrial (SA) node**, a group of specialized fibers located in the upper posterior wall of the right atrium. These specialized cells spontaneously depolarize at a faster rate than other myocardial muscle cells because they are especially leaky to sodium ions. Because this tissue has the fastest intrinsic firing rate, typically at a frequency of about 100 beats/min, the SA node is known as the heart's pacemaker, and the rhythm it establishes is called the *sinus rhythm*. The electrical impulse generated by the SA node spreads through both atria and reaches the **atrioventricular (AV) node**, located in the right atrial wall near the center of the heart. As the electrical impulse spreads through the atria, the atrial myocardium is signaled to contract.

The AV node conducts the electrical impulse from the atria into the ventricles. The impulse is delayed by about 0.13 s as it passes through the AV node, and then it enters the AV bundle. This delay is important because it allows blood from the atria to completely empty into the ventricles to maximize ventricular filling before the ventricles contract. While most blood moves passively from the atria to the ventricles, active contraction of the atria (sometimes called the "atrial kick") completes the process. The AV bundle travels along the ventricular septum and then sends right and left bundle branches into the respective ventricles. These branches send the impulse toward the apex of the heart and then outward. Each bundle branch subdivides into many smaller ones that spread throughout the entire ventricular wall. These terminal branches of the AV bundle are the Purkinje fibers. They transmit the impulse through the ventricles approximately six times faster than through the rest of the cardiac conduction system. This rapid conduction allows all parts of the ventricle to contract at virtually the same time.

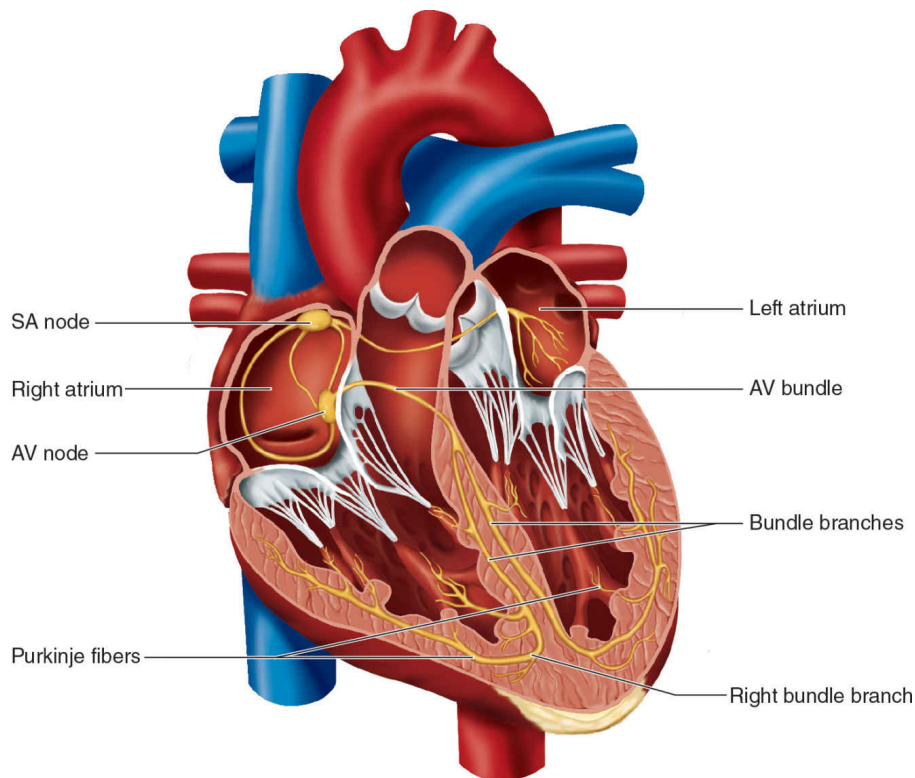


FIGURE 6.5 The specialized conduction system of the heart.

Occasionally, chronic problems develop within the cardiac conduction system, hampering its ability to maintain appropriate sinus rhythm throughout the heart. In such cases, an artificial pacemaker can be surgically installed. This small, battery-operated electrical stimulator, usually implanted under the skin, has tiny electrodes attached to the right ventricle. For example, in a condition called *AV block*, the SA node creates an impulse, but the impulse is blocked at the AV node and cannot reach the ventricles, resulting in the heart rate's being controlled by the intrinsic firing rate of the pacemaker cells in the ventricles (closer to 40 beats/min). The artificial pacemaker takes over the role of the disabled AV node, supplying the needed impulse and thus controlling ventricular contraction.

Extrinsic Control of Heart Rate and Rhythm

Although the heart initiates its own electrical impulses (intrinsic control), both the heart rate and force of contraction can be altered.

Under normal conditions, this is accomplished primarily through three extrinsic systems:

- The parasympathetic nervous system
- The sympathetic nervous system
- The endocrine system (hormones)

Although an overview of these systems' effects is offered here, they are discussed in more detail in [chapters 3 and 4](#).

The parasympathetic system, a branch of the autonomic nervous system, originates centrally in a region of the brain stem called the medulla oblongata and reaches the heart through the vagus nerve (cranial nerve X). The vagus nerve carries impulses to the SA and AV nodes, and when stimulated it releases acetylcholine, which causes hyperpolarization of the conduction cells. The result is a slower spontaneous depolarization and a decrease in heart rate. At rest, parasympathetic system activity predominates and the heart is said to have “vagal tone.” Recall that, in the absence of vagal tone, intrinsic heart rate would be approximately 100 beats/min, but the normal resting adult heart rate is typically 60 to 80 beats/min. The vagus nerve has a depressant effect on the heart: It slows impulse generation and conduction and thus decreases the heart rate. Maximal vagal stimulation can decrease the heart rate to as low as 20 beats/min. The vagus nerve also decreases the force of cardiac muscle contraction.

The sympathetic nervous system, the other branch of the autonomic system, has opposite effects. Sympathetic stimulation increases the rate of depolarization of the SA node as well as conduction speed, and thus heart rate. Maximal sympathetic stimulation can increase the heart rate to 250 beats/min. Sympathetic input also increases the force of contraction of the ventricles. Sympathetic control predominates during times of physical or emotional stress when the heart rate is greater than 100 beats/min. The parasympathetic system dominates when heart rate is less than 100. Thus, when exercise begins, or if exercise is at a low intensity, heart rate first increases due to withdrawal of vagal tone, then increases further due to sympathetic activation, as shown in [figure 6.6](#).

The third extrinsic influence, the endocrine system, exerts its effect through two hormones released by the adrenal medulla: norepinephrine and epinephrine (see [chapter 4](#)). These hormones are also known as catecholamines. Like the norepinephrine released as a neurotransmitter by the sympathetic nervous system, circulating norepinephrine and epinephrine stimulate the heart, increasing its rate and contractility. In fact, release of these hormones from the adrenal medulla is triggered by sympathetic stimulation during times of stress, and their actions prolong the sympathetic response.

Normal resting heart rate (RHR) is defined as between 60 and 100 beats/min. With extensive endurance training (over months to years), RHR can decrease to 35 beats/min or less. A RHR as low as 28 beats/min has been observed in a world-class long-distance runner. While it has been widely accepted that these lower training-induced RHRs result primarily from increased parasympathetic stimulation (vagal tone), the actual mechanisms responsible for this training-induced sinus bradycardia remain an area of much debate (see Research Perspective 6.1).

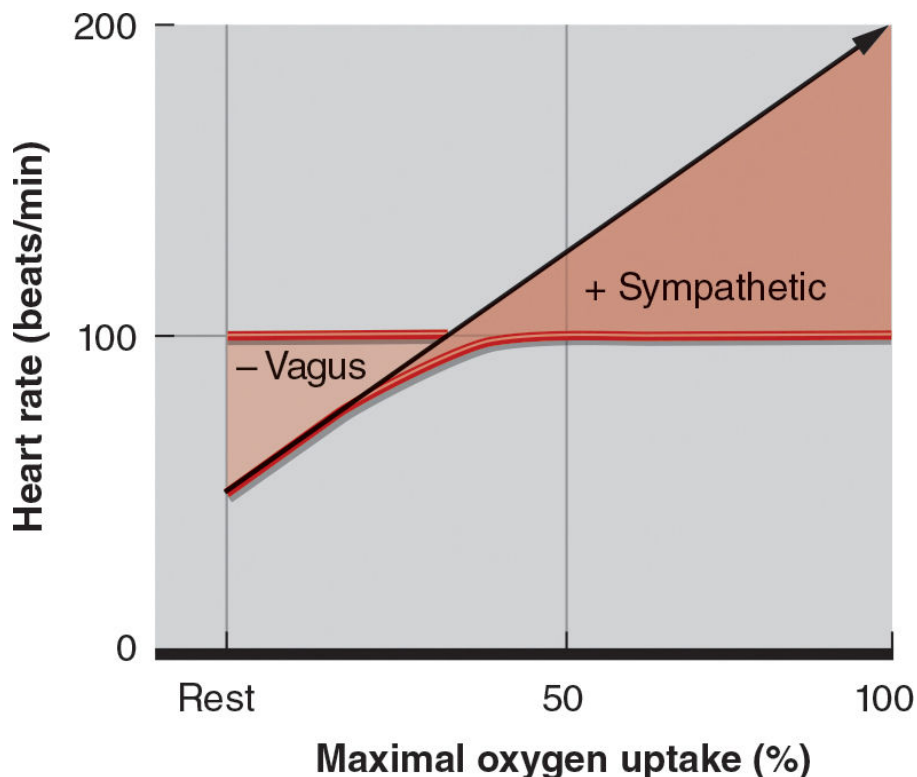


FIGURE 6.6 Relative contribution of sympathetic and parasympathetic nervous systems to the rise in heart rate from rest to exercise of increasing intensity.

Adapted from Rowell (1993).

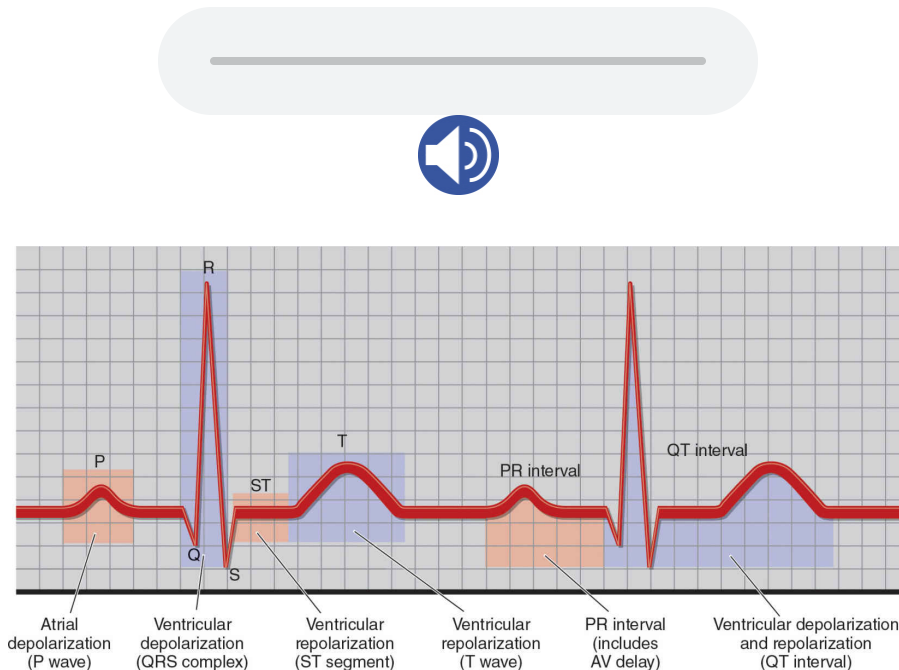


FIGURE 6.7 A graphic illustration of the various phases of the resting electrocardiogram.

Electrocardiogram

The electrical activity of the heart can be recorded to monitor cardiac changes or diagnose potential cardiac problems. Because body fluids contain electrolytes, they are good electrical conductors. Electrical impulses generated in the heart are conducted through body fluids to the skin, where they can be amplified, detected, and printed out on an **electrocardiograph**. This printout is called an **electrocardiogram (ECG)**. A standard ECG is recorded from 10 electrodes placed in specific anatomical locations. These 10 electrodes correspond to 12 leads that represent different views of the heart. Three basic components of the ECG represent important aspects of cardiac function (figure 6.7):

- The P wave
- The QRS complex
- The T wave

The P wave represents atrial depolarization and occurs when the electrical impulse travels from the SA node through the atria to the AV node. The QRS complex represents ventricular depolarization and occurs as the impulse spreads from the AV bundle to the **Purkinje fibers** and through the ventricles. The T wave represents ventricular repolarization. Atrial repolarization cannot be seen, because it occurs during ventricular depolarization (QRS complex).

It is important to remember that an ECG measures only the electrical activity of the heart and does not provide any information about its function as a pump. Electrocardiograms are often obtained at rest, then again during exercise as clinical diagnostic tests of cardiac function. As exercise intensity increases, the heart must beat faster and work harder to deliver more blood to active muscles. Indications of coronary artery disease not evident at rest may show up on the ECG as the strain on the heart increases.

In Review

- The atria serve primarily as filling chambers, receiving blood from the veins; the ventricles are the primary pumps that eject blood from the heart.
- Because the left ventricle must produce more force than other chambers to pump blood throughout the systemic circulation, its myocardial wall is thicker.
- Cardiac tissue is capable of spontaneous rhythmicity and has its own specialized conduction system made up of myocardial fibers that serve specialized functions.
- Because it has the fastest inherent rate of depolarization, the SA node is normally the heart's pacemaker.
- Heart rate and force of contraction can be altered by the autonomic nervous system (sympathetic and parasympathetic) and the endocrine system through circulating catecholamines (epinephrine and norepinephrine).
- Electrocardiograms are often obtained during exercise as clinical diagnostic tests of cardiac function. Indications of coronary artery disease not evident at rest may show up on the ECG as the strain on the heart increases.
- The ECG provides no information about the pumping capacity of the heart, only its electrical activity.

Cardiac Arrhythmias

Occasionally, disturbances in the normal sequence of cardiac events can lead to an irregular heart rhythm, called an arrhythmia. These disturbances vary in degree of seriousness. Bradycardia and tachycardia are two types of arrhythmias. **Bradycardia** is defined as a RHR lower than 60 beats/min, whereas **tachycardia** is defined as a resting rate greater than 100 beats/min. With these arrhythmias, the sinus rhythm is normal, but the rate is altered. In extreme cases, bradycardia or tachycardia can affect maintenance of blood pressure. Symptoms of both arrhythmias include fatigue, dizziness, light-headedness, and fainting. Tachycardia can sometimes be sensed as palpitations or a racing pulse. Interestingly, highly trained endurance athletes also develop a resting bradycardia, an advantageous adaptation. This adaptation should not be confused with pathological causes of bradycardia. Nor should the elevated heart rate during exercise be confused with a tachycardia indicative of underlying disease or dysfunction.

Other arrhythmias may also occur. For example, **premature ventricular contractions (PVCs)**, which result in the feeling of skipped or extra beats, are relatively common and result from impulses originating outside the SA node. Atrial flutter, in which the atria depolarize at rates of 200 to 400 beats/min, and atrial fibrillation, in which the atria depolarize in a rapid and uncoordinated manner, are more serious arrhythmias that may cause ventricular filling problems. **Ventricular tachycardia**, defined as three or more consecutive PVCs, is a very serious arrhythmia that compromises the pumping capacity of the heart and can lead to **ventricular fibrillation**, in which depolarization of the ventricular tissue is random and uncoordinated. When this happens, the heart is extremely inefficient, and little or no blood is pumped out of the heart. Under such conditions, the use of a defibrillator to shock the heart back into a normal sinus rhythm must occur within minutes if the victim is to survive.

The Cardiac Cycle

The **cardiac cycle** includes all the mechanical and electrical events that occur during one heartbeat. In mechanical terms, all heart chambers undergo a relaxation phase (diastole) and a contraction

phase (systole). During diastole, the chambers fill with blood. During systole, the ventricles contract and expel blood into the aorta and pulmonary arteries. The diastolic phase is approximately twice as long as the systolic phase. Consider an individual with a heart rate of 74 beats/min. At this heart rate, the entire cardiac cycle takes 0.81 s to complete (60 s divided by 74 beats). Of the total cardiac cycle at this rate, diastole accounts for 0.50 s, or 62% of the cycle, and systole accounts for 0.31 s, or 38%. As the heart rate increases, these time intervals shorten proportionately.

RESEARCH PERSPECTIVE 6.1

The Debate Surrounding Exercise Training–Induced Reductions in Heart Rate

It is well established that endurance exercise training lowers resting heart rate. Sinus bradycardia (a slow but otherwise normal heart rate) is evident in endurance athletes, whose resting heart rates can be half of that of their sedentary age-matched peers. However, despite substantial research, the mechanisms responsible for this training-induced reduction in resting heart rate remain an area of much debate.^{1,2}

Two primary hypotheses have been proposed to explain exercise training–induced reductions in heart rate. The first, termed the autonomic neural hypothesis, posits that reductions in heart rate result from a shift in autonomic neural balance (sympathetic versus parasympathetic influences) toward increased parasympathetic activity. The second, referred to as the intrinsic rate hypothesis, suggests that changes in inherent cardiac pacemaker rate (i.e., rate of spontaneous depolarization of the sinoatrial [SA] node cells) govern reductions in heart rate following training.

Evidence put forth in support of the autonomic neural hypothesis is largely derived from the indirect evaluation of cardiac autonomic regulation from changes in heart rate variability or systemic pharmacological interventions that were not selective to cardiac function. For this hypothesis to be correct, the selective surgical elimination of cardiac autonomic innervation should prevent reductions in resting heart rate following exercise training. This has been demonstrated in an experimental dog model as well as in human cardiac transplant patients. That is, surgical elimination of all cardiac innervation completely prevented the exercise training–induced bradycardia. These data provide direct support for the autonomic neural hypothesis because they demonstrate that intact autonomic innervation of the heart (specifically, parasympathetic) is necessary for exercise training to result in reductions in resting heart rate.

However, equally convincing data have been set forth in support of the intrinsic rate hypothesis. With this alternate hypothesis, training bradycardia results not from increased vagal tone but instead from an electrical remodeling of the SA node itself. In support of this hypothesis, exercise-trained rodents display downregulation of cardiac ion channels, directly changing the function of the pacemaker of the heart, the SA node. Furthermore, prevention of ion channel downregulation abolished the difference in heart rate between trained and untrained animals. When considered collectively, these data provide support for the concept that alterations in SA node function mediate training-induced reductions in heart rate.

Regardless of whether the autonomic neural hypothesis or the intrinsic rate hypothesis, or a synthesis of the two, is ultimately proven correct, this ongoing debate highlights the critical importance of the scientific method in reaching these conclusions. That is, experiments must be conducted to test a specific hypothesis, and these experiments must be well controlled and adequately powered in order for the results to advance scientific discussion and discovery.

Refer to the normal ECG in [figure 6.7](#). One cardiac cycle spans the time between one systole and the next. Ventricular contraction (systole) begins during the QRS complex and ends in the T wave. Ventricular relaxation (diastole) occurs during the T wave and continues until the next contraction. Although the heart is continually working, it spends slightly more time in diastole (~2/3 of the cardiac cycle) than in systole (~1/3 of the cardiac cycle).

The pressure inside the heart chambers rises and falls during each cardiac cycle. When the atria are relaxed, blood from the venous circulation fills the atria. About 70% of the blood filling the atria during this time passively flows directly through the mitral and tricuspid valves into the ventricles. When the atria contract, the atria push the remaining 30% of their volume into the ventricles.

During ventricular diastole, the pressure inside the ventricles is low, allowing the ventricles to passively fill with blood. As atrial contraction provides the final filling volume of blood, the pressure inside the ventricles increases slightly. As the ventricles contract, pressure inside the ventricles rises sharply. This increase in ventricular pressure forces the atrioventricular valves (i.e., tricuspid and mitral valves) closed, preventing any backflow of blood from the ventricles to the atria. The closing of the atrioventricular valves

results in the first heart sound. Then, when ventricular pressure exceeds the pressure in the pulmonary artery and the aorta, the pulmonary and aortic valves open, allowing blood to flow into the pulmonary and systemic circulations, respectively. Following ventricular contraction, pressure inside the ventricles falls and the pulmonary and aortic valves close. The closing of these valves corresponds to the second heart sound. The two sounds together, the result of valves closing, results in the typical “lub, dub” heard through a stethoscope during each heartbeat.

The interactions of the various events that take place during one cardiac cycle are illustrated in [figure 6.8](#), called a Wiggers diagram after the physiologist who created it. The diagram integrates information from the electrical conduction signals (ECG), heart sounds from the heart valves, pressure changes within the heart chambers, and left ventricular volume.

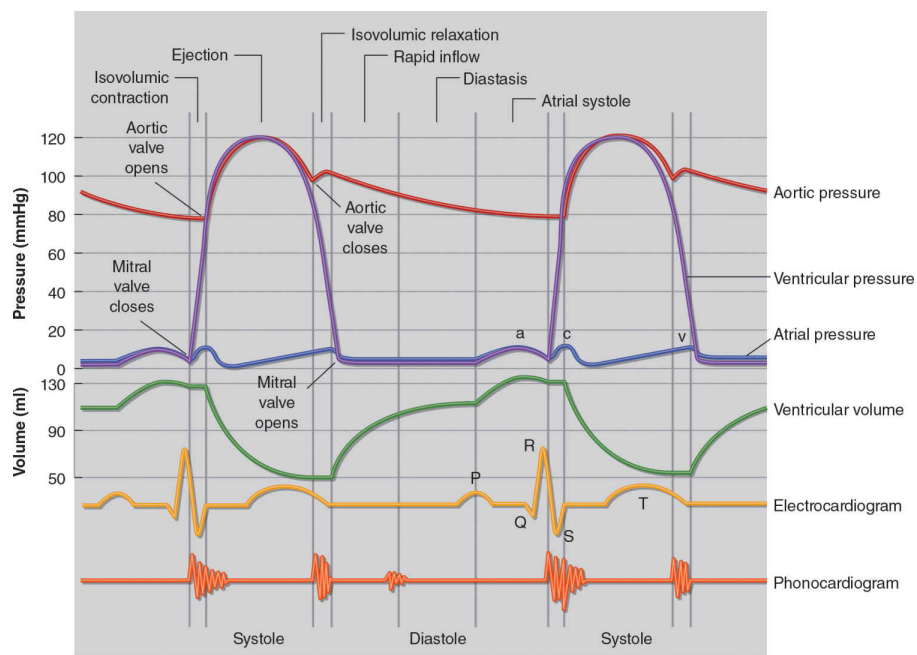


FIGURE 6.8 The Wiggers diagram, illustrating the events of the cardiac cycle for left ventricular function. Integrated into this diagram are the changes in left atrial and ventricular pressure, aortic pressure, ventricular volume, electrical activity (electrocardiogram), and heart sounds.



Determinants of Cardiac Output

The heart's primary function is as a pump. The volume of blood pumped by the heart each minute governs blood flow to living tissues and, in the case of working muscle, is a key determinant of exercise performance.

Stroke Volume

During systole, most, but not all, of the blood in the ventricles is ejected. This volume of blood pumped during one beat (contraction) is the **stroke volume (SV)**. This is depicted in [figure 6.9a](#). To understand SV, consider the amount of blood in the ventricle before and after contraction. At the end of diastole, just before contraction, the ventricle has finished filling. The volume of blood it now contains is called the **end-diastolic volume (EDV)**. At rest in a normal healthy adult, this value is approximately 100 ml. At the end of systole, just after the contraction, the ventricle has completed its ejection phase, but not all the blood is pumped out of the heart. The volume of blood remaining in the ventricle is called the **end-systolic volume (ESV)** and is approximately 40 ml under resting conditions. Stroke volume is the volume of blood that was ejected and is merely the difference between the volume of the filled ventricle and the volume remaining in the ventricle after contraction. So, SV is simply the difference between EDV and ESV; that is, $SV = EDV - ESV$ (example: $SV = 100 \text{ ml} - 40 \text{ ml} = 60 \text{ ml}$).

Ejection Fraction

The fraction of the blood pumped out of the left ventricle in relation to the amount of blood that was in the ventricle before contraction is called the **ejection fraction (EF)**. Ejection fraction is determined by dividing the SV by EDV ($60 \text{ ml} / 100 \text{ ml} = 60\%$), as in [figure 6.9b](#). The EF, generally expressed as a percentage, averages about 60% at rest in healthy, active young adults. Thus, 60% of the blood in the ventricle at the end of diastole is ejected with the next contraction,

and 40% remains in the ventricle. Ejection fraction is often used clinically as an index of the pumping ability of the heart.

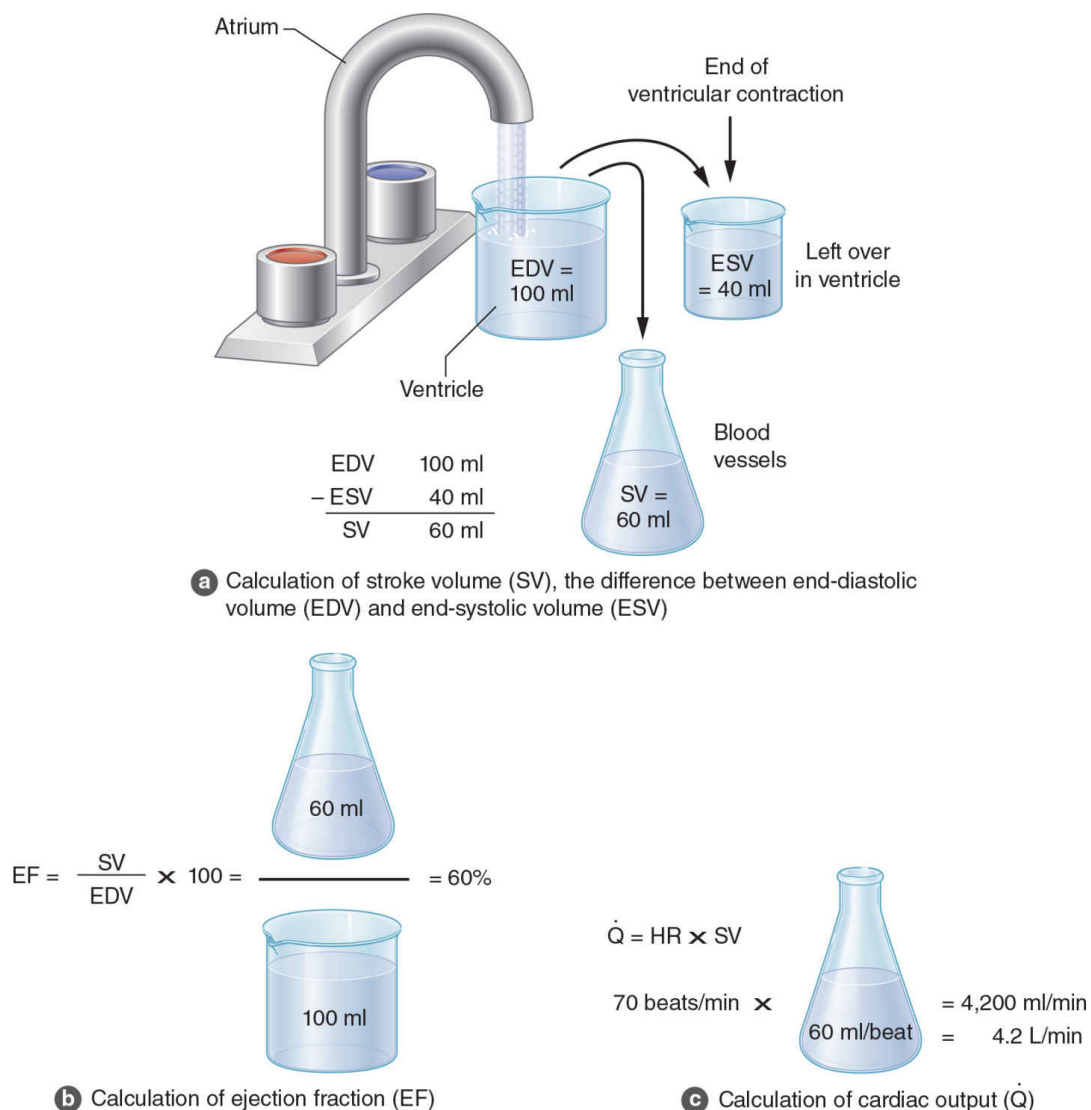


FIGURE 6.9 Calculations of stroke volume, ejection fraction, and cardiac output based on volumes of blood flowing into and out of the heart.

Cardiac Output

Cardiac output ($\dot{Q}$), as shown in figure 6.9c, is the total volume of blood pumped by the ventricle per minute, the product of heart rate (HR) and SV. The SV at rest in the standing posture averages between 60 and 80 ml of blood in most adults. Thus, at a RHR of 70 beats/min, the resting cardiac output will vary between 4.2 and 5.6 L/min. The average adult body contains about 5 L of blood, so this

means that the equivalent of our total blood volume is pumped through our hearts about once every minute.

Pumping Action of the Heart During Exercise

As described earlier in this chapter, the myocardium has to contract as if it were a single unit in order to efficiently pump blood. For that reason, myocardial cells are anatomically interconnected end to end by intercalated disks that anchor the individual cells together so that they do not pull apart during contraction. This better allows the heart to contract as one unit, often called a functional syncytium.

VIDEO 6.1 Presents Ben Levine on torsional contraction of the heart muscle and its role in ventricular filling.



During intense exercise when heart rates are high, the time available between contractions for diastolic filling is very short. Yet complete filling of the left ventricle must occur in order to appropriately increase cardiac output. The heart actually uses the increased contractility that occurs during exercise to enhance left ventricle filling, a process called *torsional contraction*. As the heart beats, contraction and relaxation of the atria and the ventricles create a twisting and untwisting action, similar to wringing out a towel. During systole (contraction), the heart twists gradually, storing energy and compressing the springlike titin molecules in the sarcomere (see [chapter 1](#) for a description of titin's similar role in skeletal muscle). When the aortic valve closes, the ventricle abruptly untwists. This recoil creates a 1 to 2 mmHg pressure difference

between the base (top) and apex (bottom) of the heart, which pulls blood from the atrium, across the mitral valve, and into the ventricle.

RESEARCH PERSPECTIVE 6.2

Can Too Much Exercise Be Bad for Your Heart?

It is well known that habitual physical activity reduces cardiovascular disease risk and that exercise dose is also important; higher physical activity levels further reduce mortality risk, and the most active individuals demonstrate the highest overall life expectancy. However, few studies have included individuals engaging in lifelong *high-intensity* endurance exercise. This is an important gap in our knowledge, since recent evidence suggests that such intense exercise may paradoxically *increase* cardiovascular risk.

Intense exercise performed regularly elicits structural, functional, and electrical cardiac adaptations, collectively known as the athlete's heart. As elegantly reviewed by Eijssvogels and colleagues,⁸ these adaptations may also impart deleterious effects. In response to exercise training, all four chambers of the heart enlarge. Although this adaptation facilitates exercise performance, it may also have adverse cardiac effects. For example, atrial fibrillation becomes more common, potentially resulting from increased vagal tone and left atrial size. Further, right ventricular wall stress is increased, possibly due to exercise-induced increases in pulmonary artery systolic pressure. Together, these changes in heart structure may hasten cardiac disease in susceptible individuals.

In addition to structural adaptations, exercise also acutely increases circulating biomarkers for cardiovascular disease, including creatine kinase, cardiac troponin, and B-type natriuretic peptide. Although the source of these circulating molecules remains unclear, the increases likely result from both skeletal muscle damage and stress-activated cardiac muscle. These increases may be cause for concern because prolonged exercise reduces ventricular function and acutely injures cardiac muscle, resulting in cardiac fatigue. Myocardial fibrosis (a buildup of scar tissue in the cardiac muscle or valves) has also been documented in some lifelong endurance athletes. The interrelation between increases in cardiac biomarkers, reductions in ventricular function, and cardiac fibrosis may impart increased risk, though the precise mechanisms remain incompletely understood.

While the possibility that lifelong intense endurance exercise may increase cardiac risk cannot be ignored, for the majority of the population, the evidence supporting the beneficial cardiovascular outcomes attributed to exercise and physical activity is overwhelming. It is these health benefits that led to the development of strategies to increase physical activity, with very specific guidelines put forth by both the American College of Sports Medicine and the American Heart Association. Unfortunately, the majority of Americans

fail to meet these exercise criteria, putting themselves at an ever-increased risk of cardiovascular disease.

The torsion during systole stores energy that is then released during isovolumetric relaxation (the period during the cardiac cycle after contraction when all of the valves are closed and the myocardium is relaxing) to generate the diastolic suction that allows enhanced atrial filling during exercise. This twisting action is enhanced almost threefold during exercise and assists in efficient ventricular filling. This left ventricle twisting and rapid untwisting, inducing diastolic suction in the ventricle, is called *dynamic relaxation*.¹⁰ Cardiac twisting mechanics are improved with exercise training and reduced by detraining.⁷

In Review

- The electrical and mechanical events that occur in the heart during one heartbeat make up one cardiac cycle. A Wiggers diagram depicts the intricate timing of these events.
- Cardiac output, the volume of blood pumped by each ventricle per minute, is the product of HR and SV.
- Not all of the blood in the ventricles is ejected during systole. The ejected volume is the SV, while the percentage of blood pumped with each beat is the EF.
- To calculate SV, EF, and cardiac output:

$$\text{SV (ml/beat)} = \text{EDV} - \text{ESV}$$

$$\text{EF (\%)} = (\text{SV}/\text{EDV}) \times 100$$

$$\dot{Q} \text{ (L/min)} = \text{HR} \times \text{SV}$$

- As the heart beats, contraction and relaxation of the atria and the ventricles create a twisting and untwisting action that allows the ventricles to fill even at high heart rates.

Understanding the electrical and mechanical activity of the heart provides a basis for understanding the cardiovascular system, but the heart is only one part of this system. In addition to this pump, the cardiovascular system contains an intricate network of tubes that serve as a delivery system carrying the blood to all body tissues.

Vascular System

The vascular system contains a series of vessels that transport blood from the heart to the tissues and back: the arteries, arterioles, capillaries, venules, and veins.

Arteries are large, muscular, elastic conduit vessels for transporting blood away from the heart to the arterioles. The aorta is the largest artery, transporting blood from the left ventricle to all regions of the body as it eventually branches into smaller and smaller arteries, finally branching into arterioles. The **arterioles** are the site of greatest control of the circulation by the sympathetic nervous system, so arterioles are sometimes called *resistance vessels*. Arterioles are heavily innervated by the sympathetic nervous system and are the main site of control of blood flow to specific tissues.

From the arterioles, blood enters the **capillaries**, the narrowest and simplest vessels in terms of their structure, with walls only one cell thick. Virtually all exchange between the blood and the tissues occurs at the capillaries. Blood leaves the capillaries to begin the return trip to the heart in the **venules**, and the venules form larger vessels—the **veins**. The vena cava is the great vein transporting blood back to the right atrium from all regions of the body above (superior vena cava) and below (inferior vena cava) the heart.

Blood Pressure

Blood pressure is the pressure exerted by the blood on the arterial walls. It is expressed by two numbers: the **systolic blood pressure (SBP)** and the **diastolic blood pressure (DBP)**. The higher number is the SBP; it represents the highest pressure in the artery that occurs during ventricular systole. Ventricular contraction pushes the blood through the arteries with tremendous force, and that force exerts high pressure on the arterial walls. The lower number is the DBP and represents the lowest pressure in the artery, corresponding to ventricular diastole when the ventricle is filling.

Mean arterial pressure (MAP) represents the average pressure exerted by the blood as it travels through the arteries. Since diastole takes about twice as long as systole in a normal cardiac cycle, MAP can be estimated from the DBP and SBP as follows:

$$\text{MAP} = 2/3 \text{ DBP} + 1/3 \text{ SBP}$$

Alternatively,

$$\text{MAP} = \text{DBP} + [0.333 \times (\text{SBP} - \text{DBP})]$$

(SBP – DBP) is also called pulse pressure.

To illustrate, with a normal resting blood pressure of 120 mmHg over 80 mmHg, the $\text{MAP} = 80 + [0.333 \times (120 - 80)] = 93 \text{ mmHg}$.

General Hemodynamics

The cardiovascular system is a continuous closed-loop system. Blood flows through this closed loop because of the pressure gradient that exists between the arterial and venous sides of the circulation. To understand regulation of blood flow to the tissues, it is necessary to understand the intricate relationship between pressure, blood flow, and resistance.

In order for blood to flow through a vessel, there must be a pressure difference from one end of the vessel to the other end. Blood will flow from the region of the vessel with high pressure to the region of the vessel with low pressure. Alternatively, if there is no pressure difference across the vessel, there is no driving force and therefore no blood flow. In the circulatory system, the MAP in the aorta is approximately 100 mmHg at rest, and the pressure in the right atrium is very close to 0 mmHg. Therefore, the pressure difference across the entire circulatory system is $100 \text{ mmHg} - 0 \text{ mmHg} = 100 \text{ mmHg}$.

The reason for the pressure differential from the arterial to the venous circulation is that the blood vessels themselves provide resistance to blood flow. The resistance that the vessel provides is largely dictated by the properties of the blood vessel and the blood itself. These properties include the length and radius of the blood vessel and the viscosity or thickness of the blood flowing through the vessel. Resistance to flow can be calculated as

$$\text{Resistance} = \eta \times L / r^4$$

where η is the viscosity (thickness) of the blood, L is the length of the vessel, and r is the radius of the vessel, which is raised to the fourth power. Blood flow is proportional to the pressure difference across

the system and is inversely proportional to resistance. This relationship can be illustrated by the following equation:

$$\text{Blood flow} = \Delta\text{pressure} / \text{resistance}$$

Notice that blood flow can increase by either an increase in the pressure difference ($\Delta\text{pressure}$), a decrease in resistance, or a combination of the two. Altering resistance to control blood flow is much more advantageous because very small changes in blood vessel radius equate to large changes in resistance. This is due to the fourth-power mathematical relationship between vascular resistance and vessel radius.

Changes in vascular resistance are largely due to changes in the radius or diameter of the vessels, since the viscosity of the blood and the length of the vessels do not change significantly under normal conditions. Therefore, regulation of blood flow to organs is accomplished by small changes in vessel radius through **vasoconstriction** and **vasodilation**. This allows the cardiovascular system to divert blood flow to the areas where it is needed most.

As mentioned earlier, most resistance to blood flow occurs in the arterioles. [Figure 6.10](#) shows the blood pressure changes across the entire vascular system. The arterioles are responsible for ~70% to 80% of the drop in MAP across the entire cardiovascular system. This is important because small changes in arteriole radius can greatly affect the regulation of mean arteriole pressure and the local control of blood flow. At the capillary level, changes due to systole and diastole are no longer evident, and the flow is smooth (laminar) rather than turbulent.

In Review

- Systolic blood pressure is the highest pressure within the vascular system, whereas DBP is the lowest pressure.
- Mean arterial pressure is the average pressure on the vessel walls during a cardiac cycle; however, it is not the mathematical mean of SBP and DBP because diastole takes about twice as long as systole.
- In terms of the entire cardiovascular system, cardiac output is the blood flow to the entire system, the $\Delta\text{pressure}$ is the difference between aortic pressure when

blood leaves the heart and venous pressure when blood returns to the heart, and resistance is the impedance to blood flow from the blood vessels.

- Blood flow is mainly controlled by small changes in blood vessel (arteriole) radius that greatly change resistance.
-

Distribution of Blood

Distribution of blood to the various body tissues varies considerably depending on the immediate needs of a specific tissue compared with those of other areas of the body. As a general rule, the most metabolically active tissues receive the greatest blood supply. At rest under normal conditions, the liver and kidneys combine to receive approximately half of the cardiac output, while resting skeletal muscles receive only about 15% to 20%.

During exercise, blood is redirected to the areas where it is needed most. During heavy endurance exercise, contracting muscles may receive 80% or more of the blood flow, and flow to the liver and kidneys decreases. This redistribution, along with increases in cardiac output (discussed in [chapter 8](#)), allows up to 25 times more blood flow to active muscles (see [figure 6.11](#)).

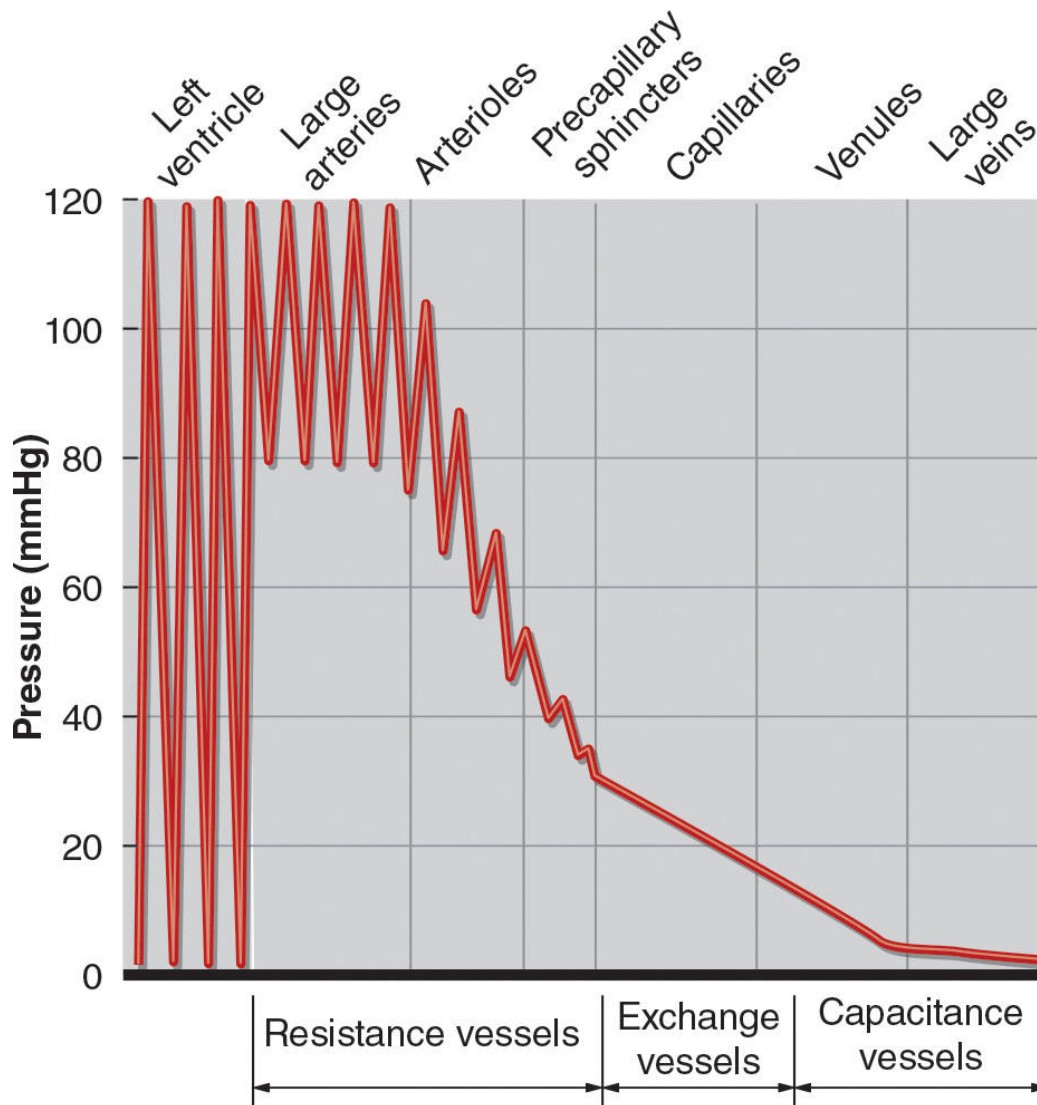


FIGURE 6.10 Pressure changes across the systemic circulation. Notice the large pressure drop that occurs across the arteriole portion of the system.

Alternatively, after one eats a big meal, the digestive system receives more of the available cardiac output than when the digestive system is empty. Along the same lines, during increasing environmental heat stress, skin blood flow increases to a greater extent as the body attempts to maintain normal temperature. The cardiovascular system responds accordingly to redistribute blood, whether it is to the exercising muscle to match metabolism, for digestion, or to facilitate thermoregulation. These changes in the distribution of cardiac output are controlled by the sympathetic nervous system, primarily by increasing or decreasing the diameter

of the arterioles providing blood flow to the given tissue or organ. Arterioles have a strong muscular wall that can significantly alter vessel diameter, are highly innervated by sympathetic nerves, and have the capacity to respond to local control mechanisms.

Intrinsic Control of Blood Flow

Intrinsic control of blood distribution refers to the ability of the local tissues to dilate or constrict the arterioles that serve them and alter regional blood flow depending on the immediate needs of those tissues. With exercise and the increased metabolic demand of the exercising skeletal muscles, the arterioles undergo intrinsic vasodilation, opening up to allow more blood to enter the highly active tissue.

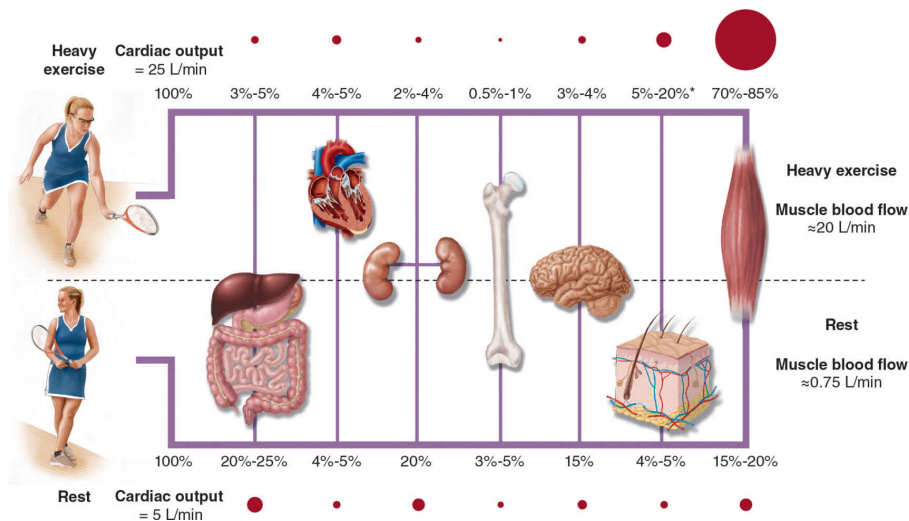


FIGURE 6.11 Distribution of cardiac output at rest and maximal exercise.

*Depends on ambient and body temperatures.

Reprinted by permission from P.O. Åstrand et al., *Textbook of Work Physiology: Physiological Bases of Exercise*, 4th ed. (Champaign, IL: Human Kinetics, 2003), 143.



There are essentially three types of intrinsic control of blood flow. *Metabolic regulation*, in response to an increased oxygen demand, is

the strongest stimulus for the release of local vasodilating chemicals. As oxygen uptake by metabolically active tissues increases, available oxygen is diminished. Local arterioles dilate to allow more blood to perfuse the area, delivering more oxygen. Other chemical changes that can stimulate increased blood flow are decreases in other nutrients and increases in by-products (carbon dioxide, K^+ , H^+ , lactic acid) or inflammatory molecules.

Second, many dilator substances can be produced within the endothelium (inner lining) of arterioles that can initiate vasodilation in the vascular smooth muscle of those arterioles (*endothelium-mediated vasodilation*). These substances include nitric oxide (NO), prostaglandins, and endothelium-derived hyperpolarizing factor (EDHF). These endothelium-derived vasodilators are important regulators of blood flow at rest and during exercise in humans. Additionally, acetylcholine and adenosine have been proposed as potential vasodilators for the increase in muscle blood flow during exercise.

Third, pressure changes within the vessels themselves can also cause vasodilation and vasoconstriction. This is referred to as the *myogenic response*. The vascular smooth muscle contracts in response to an increase in pressure across the vessel wall and relaxes in response to a decrease in pressure across the vessel wall. [Figure 6.12](#) illustrates the three types of intrinsic control of vascular tone.

Extrinsic Neural Control

The concept of intrinsic local control explains redistribution of blood within an organ or tissue; however, the cardiovascular system must divert blood flow to where it is needed, beginning at a site upstream of the local environment. Redistribution at the system or organ level is controlled by neural mechanisms. This is known as **extrinsic neural control** of blood flow, because the control comes from outside the specific area (extrinsic) instead of from inside the tissues (intrinsic).

Blood flow to all body parts is regulated largely by the sympathetic nervous system. Sympathetic nerves are abundant in the circular layers of smooth muscle within the artery and arteriole walls. In most

vessels, an increase in sympathetic nerve activity causes these circular smooth muscle cells to contract, constricting blood vessels and thereby decreasing blood flow.

Under normal resting conditions, sympathetic nerves transmit impulses continuously to the blood vessels (in particular, the arterioles), keeping the vessels moderately constricted to maintain adequate blood pressure. This state of tonic vasoconstriction is referred to as *vasomotor tone*. When sympathetic stimulation increases, further constriction of the blood vessels in a specific area decreases blood flow into that area and allows more blood to be distributed elsewhere. But if sympathetic stimulation decreases below the level needed to maintain tone, constriction of vessels in the area is lessened, so the vessels passively dilate, increasing blood flow into that area. Therefore, sympathetic stimulation will cause vasoconstriction in most vessels. Blood flow can passively be increased through a lowering of the normal tonic level of sympathetic outflow.

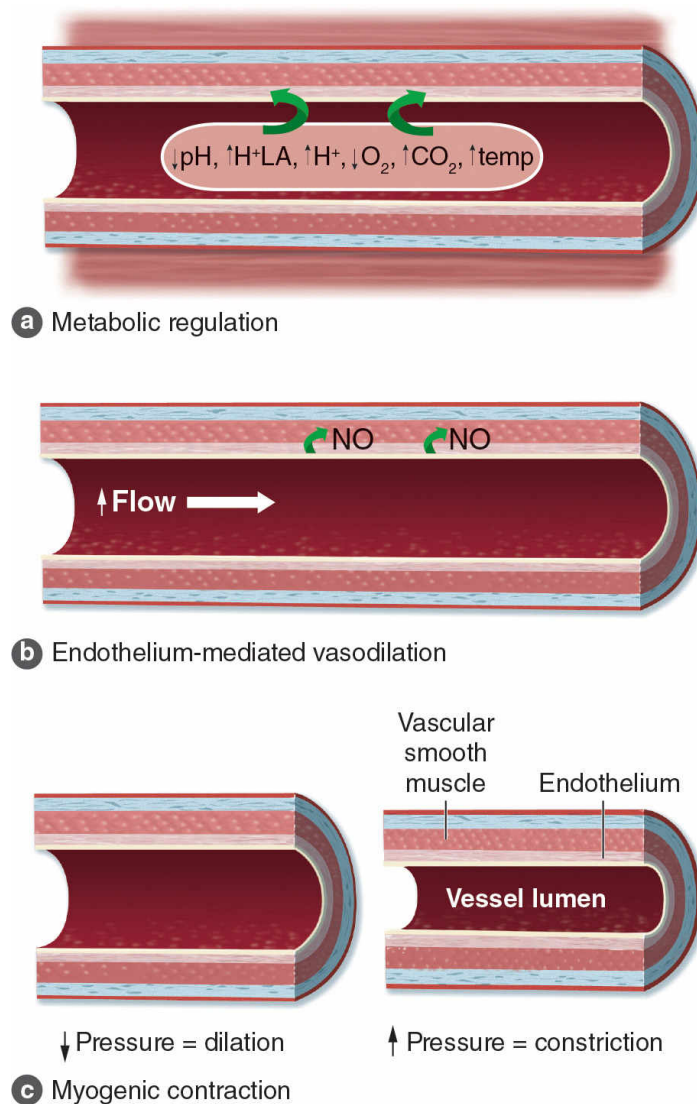


FIGURE 6.12 Intrinsic control of blood flow. Arterioles are signaled to dilate or constrict at the local level (a) by changes in the local concentration of oxygen or metabolic products, (b) by the effects of local pressure within the arterioles, and (c) by endothelium-derived factors.

Figure courtesy of Dr. Donna H. Korzick, Pennsylvania State University.



Local Control of Muscle Blood Flow

The previous two sections discuss intrinsic and extrinsic control of blood flow, mechanisms that pertain to controlling flow to all tissues

of the body. However, muscle blood flow deserves special attention because (1) contracting muscle is the hallmark response of exercise physiology and (2) unique mechanisms exist to support increased muscle blood flow. During aerobic exercise, blood flow to exercising muscle must increase to match the metabolic demand of that muscle. Enhanced oxygen delivery to exercising muscle can occur via a number of different mechanisms, including local alteration of blood flow, improved oxygen extraction at the tissue level, or both.

Exercise is accompanied by a general increase in sympathetic nerve activity, including that directed to muscle, which causes vasoconstriction. How does working muscle overcome systemic vasoconstriction and actually increase blood flow? The primary mechanism is termed **functional sympatholysis**. Vasoactive molecules released from the active skeletal muscle and endothelium have been shown to inhibit sympathetic vasoconstriction by reducing vascular responsiveness to α -adrenergic receptor activation. Endothelial cells release molecules called endothelial-derived hyperpolarizing factors (EDHFs) that make it more difficult for smooth muscle cells to constrict in response to sympathetic stimulation. For example, we now know that ATP released from the endothelium and from red blood cells can cause hyperpolarization of vascular smooth muscle cells that helps override α -adrenergic vasoconstriction. Functional sympatholysis helps optimize muscle blood flow distribution to match tissue perfusion with metabolic demand.

When oxygen availability is limited under conditions of reduced arterial O_2 content (e.g., hypoxia) or decreased perfusion pressure, the skeletal muscle arterioles dilate to compensate for reduced O_2 delivery, allowing for a greater O_2 extraction at the tissue level.⁴ This phenomenon is termed compensatory vasodilation.

In order to examine the mechanisms by which the local control of skeletal muscle blood flow is altered in exercising muscle in humans, investigators have used acute systemic hypoxia, usually by having subjects breathe air mixtures with a low O_2 content, to reduce arterial O_2 content during exercise¹¹ or have temporarily limited blood flow to the exercising muscle by partially occluding flow to the exercising limb. During submaximal hypoxic exercise, blood flow to the

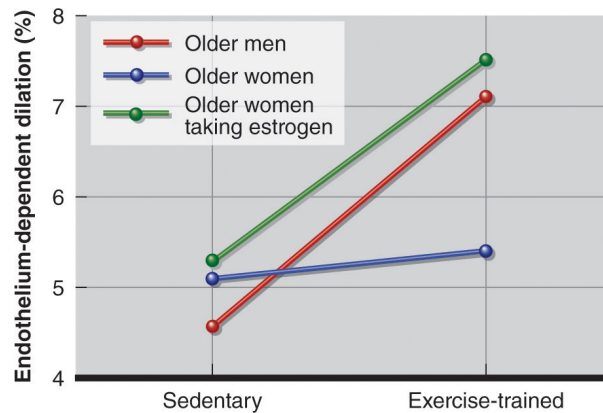
exercising muscle is the same as that seen during normoxic exercise due to the individual and combined roles of β -adrenergic receptors, adenosine, and nitric oxide (NO), as shown in [figure 6.13](#).

Interestingly, the contribution of these different dilator mechanisms can change, depending on the exercise intensity and whether blood flow to the exercising muscle is limited. For example, during low-intensity exercise under hypoxic conditions, stimulation of β -adrenergic receptors contributes to vasodilation; however, as exercise intensity increases, NO release from the endothelium contributes to a greater extent in the compensatory vasodilator response.³ The molecule adenosine can also contribute to compensatory vasodilation, especially under conditions in which blood flow is limited. At higher exercise intensities in which the muscle fibers' oxygen needs are even greater, NO and several other vasodilator molecules, including prostaglandins and adenosine triphosphate (ATP), mediate vasodilation. However, there are redundancies in these vasodilator mechanisms, such that when one is blocked or downregulated, another vasodilator can compensate and cause vasodilation.

RESEARCH PERSPECTIVE 6.3

Vascular Adaptations to Exercise Training in Postmenopausal Women

Despite recent declines in prevalence, cardiovascular disease remains the leading cause of death in the United States. Interestingly, cardiovascular mortality risk differs greatly between the sexes. The establishment of the National Institutes of Health (NIH) Office of Women's Health Research in 1990 and the passage of the NIH Revitalization Act in 1993 mandated the inclusion of women in NIH-funded research. Since the adoption of these requirements, much has been learned about the mechanisms and manifestation of cardiovascular disease in women. Yet despite these advancements, the reasons for the sex disparity in cardiovascular morbidity and mortality remain unclear.



Estrogen is required to elicit exercise training-induced improvements in vascular endothelial function in older postmenopausal women. In older men, moderate-intensity exercise training significantly improves vascular function, as assessed by endothelium-dependent dilation. This beneficial vascular adaptation to exercise did not occur in a group of postmenopausal women of similar age. However, when postmenopausal women were supplemented with estrogen, the expected improvements in vascular endothelial function following exercise training were similar to those observed in age-matched men.

Vascular aging (age-related changes in blood vessels) is considered a primary risk factor for age-associated cardiovascular disease. Endothelial dysfunction, defined as impaired endothelium-dependent dilation, is the first functional manifestation of atherosclerosis and accelerates the progression of cardiovascular disease. Lifestyle modifications, such as habitual physical activity, are commonly recommended as a first-line strategy to mitigate age-associated declines in vascular endothelial function. However, there is now an increasing awareness of potential sex differences in the beneficial effects of exercise training on vascular health in older, postmenopausal women.

In a 2006 review of the prevailing literature, researchers provided support for the notion that sex hormones, specifically estrogen, modulate the exercise training-induced improvements in vascular function in women.¹¹ That is, the reduction in estrogen that occurs during menopause consequently prevents improvements in vascular function with exercise training, as highlighted in the accompanying figure. As expected, in previously sedentary middle-aged and older men, a moderate-intensity exercise training program significantly improved vascular function. In contrast, this exercise training paradigm had no effect in older postmenopausal women. However, in sedentary postmenopausal women treated with estrogen, a moderate-intensity training program did significantly improve vascular endothelial function. These findings have been corroborated in studies using varied methodology, and provide direct evidence suggesting that estrogen is required for training-induced improvements in vascular function in postmenopausal women.

Given the rapidly aging population, effective preventive strategies to mitigate the untoward consequences of cardiovascular disease are paramount. Habitual physical activity is an important strategy for the primary

prevention of cardiovascular disease—in men and women. Yet, it is exceedingly apparent that the vasculature of older men and women responds differently to exercise training, and this differential responsiveness can likely be attributed to sex hormones, or the lack thereof. Further research is necessary to determine if additional pharmacological or nonpharmacological strategies should be coupled with exercise training in order to provide a viable therapeutic intervention that permits improvements in vascular endothelial function during exercise training in estrogen-deficient postmenopausal women.

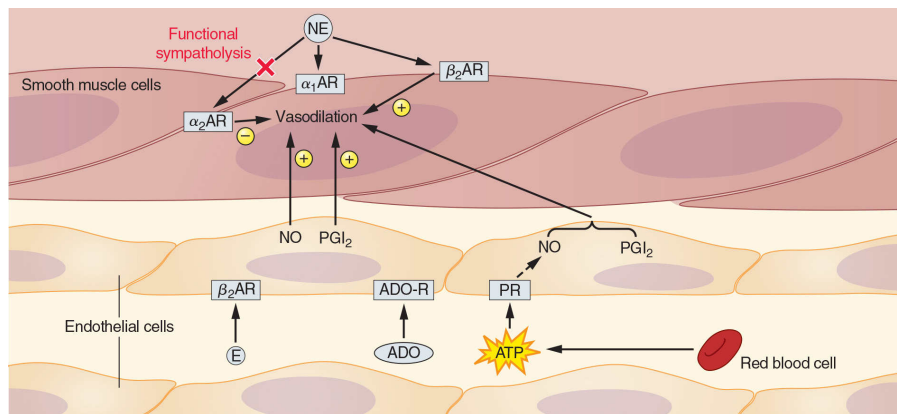


FIGURE 6.13 The proposed mechanisms for functional sympatholysis and hypoxia-induced vasodilation during exercise. During hypoxic exercise, nitric oxide (NO) is the final common pathway for the compensatory dilator response. Systemic epinephrine (E) release, acting via β -adrenergic receptors, contributes to the NO-mediated vasodilation at lower exercise intensities, but this β -adrenergic contribution decreases with increasing exercise intensity. Adenosine triphosphate (ATP) released from the red blood cell (RBC), endothelial-derived prostacyclin (PGI_2), or both remain, also stimulating NO during higher-intensity hypoxic exercise. $\alpha_1\text{AR}$, $\alpha_2\text{AR}$, and $\beta_2\text{AR}$ = α_1 -, α_2 -, and β_2 -adrenergic receptors, respectively; NE = norepinephrine; PR = purinergic receptors that are stimulated by ATP; ADO = adenosine.

Because of the biological importance of NO, these mechanisms have significant implications in clinical populations such as older individuals and patients with cardiovascular disease, in whom NO production and availability may be limited.^{5,6} For example, as humans age, there is a reduction in NO synthesis and an increase in NO breakdown, and compensatory vasodilation is blunted in healthy older humans.¹¹

Distribution of Venous Blood

While flow to tissues is controlled by changes on the arterial side of the system, most of the blood *volume* resides in the venous side of

the system. At rest, the blood volume is distributed among the vasculature as shown in [figure 6.14](#). The venous system has a great capacity to store blood volume because veins have little vascular smooth muscle and are very elastic and balloon-like. Thus, the venous system provides a large reservoir of blood available to be rapidly distributed back to the heart (venous return) and from there to the arterial circulation. This is accomplished through sympathetic stimulation of the venules and veins, which causes the vessels to constrict (venoconstriction).

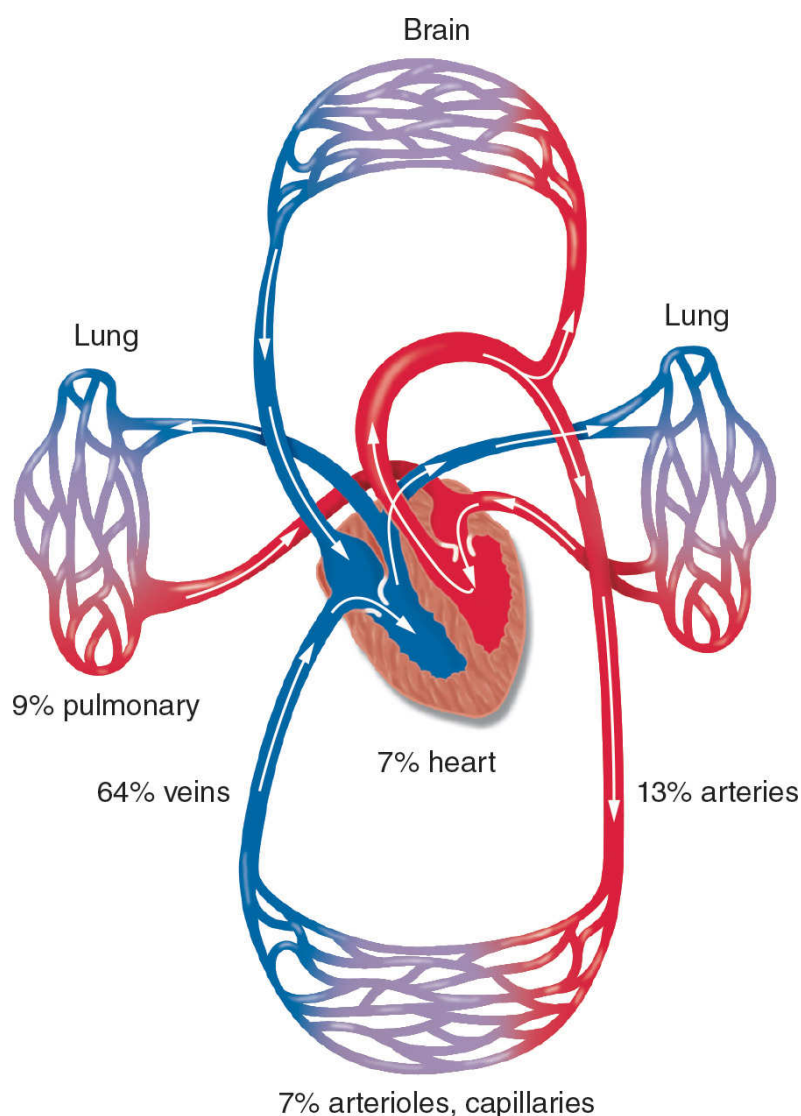


FIGURE 6.14 Blood volume distribution within the vasculature when the body is at rest.

Integrative Control of Blood Pressure

Blood pressure is controlled by reflexes. Specialized pressure sensors located in the aortic arch and the carotid arteries, called **baroreceptors**, are sensitive to changes in arterial pressure. When the pressure in these large arteries changes, afferent signals are sent to the cardiovascular control centers in the brain where autonomic reflexes are initiated, and efferent signals are sent to respond to changes in blood pressure. For example, when blood pressure is elevated, the baroreceptors are stimulated by an increase in stretch. They relay this information to the cardiovascular control center in the brain. In response to the increased pressure is an increase in vagal tone, which decreases heart rate, and a decrease in sympathetic activity to both the heart and the arterioles, which causes the arterioles to dilate. All of these adjustments serve to decrease blood pressure back to normal. In response to a decrease in blood pressure, less stretch is sensed by the baroreceptors, and the response is to increase heart rate (vagal withdrawal) and constrict arterioles (through increased sympathetic nervous activity), thus correcting the low-pressure signal.

Other specialized receptors, called **chemoreceptors** and **mechanoreceptors**, send information about the chemical environment in the muscle and the length and tension of the muscle, respectively, to the cardiovascular control centers. These receptors also modify the blood pressure response and are especially important during exercise.

Return of Blood to the Heart

Because humans spend so much time in an upright position, the cardiovascular system requires mechanical assistance to overcome the force of gravity and assist the return of venous blood from the lower parts of the body to the heart. Three basic mechanisms assist in this process:

- Valves in the veins
- The muscle pump
- The respiratory pump

The veins contain valves that allow blood to flow in only one direction, thus preventing backflow and further pooling of blood in the

lower body. These venous valves also complement the action of the skeletal **muscle pump**, the rhythmic mechanical compression of the veins that occurs during the rhythmic skeletal muscle contraction accompanying many types of movement and exercise, for example, during walking and running (figure 6.15). The muscle pump pushes blood volume in the veins back toward the heart. Finally, changes in pressure in the abdominal and thoracic cavities during breathing assist blood return to the heart by creating a pressure gradient between the veins and the chest cavity.

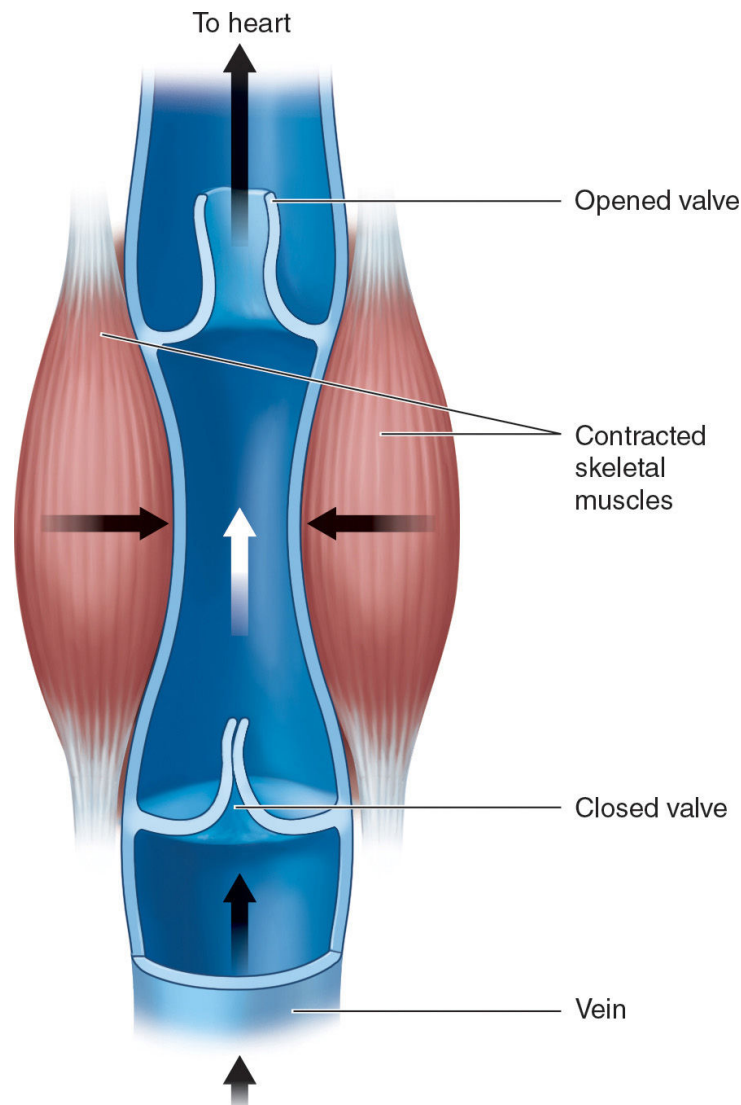


FIGURE 6.15 The muscle pump. As the skeletal muscles contract, they squeeze the veins in the legs and assist in the return of blood to the heart. Valves within the veins ensure the unidirectional flow of blood back to the heart.



In Review

- Blood is distributed throughout the body based primarily on the metabolic needs of the individual tissues. The most active tissues receive the highest blood flow.
- Skeletal muscle normally receives about 15% of the cardiac output at rest. This can increase to 80% or more during heavy endurance exercise.
- Redistribution of blood flow is controlled locally by the release of dilators from either the tissue (metabolic regulation) or the endothelium of the blood vessel (endothelium-mediated dilation). A third type of intrinsic control involves the response of the arteriole to pressure. Decreased arteriolar pressure causes vasodilation, thus increasing blood flow to the area, while increased pressure causes local constriction.
- Extrinsic neural control of blood flow distribution is accomplished by the sympathetic nervous system, primarily through vasoconstriction of small arteries and arterioles.
- During aerobic exercise, blood flow to exercising muscle must increase to match the metabolic demand of that muscle. This is accomplished by (1) functional sympatholysis (which overcomes sympathetic vasoconstriction) and (2) compensatory vasodilation (involving molecules such as adenosine and nitric oxide).
- Blood pressure is maintained under normal conditions by reflexes within the autonomic system.
- Blood returns to the heart through the veins, assisted by valves within the veins, the muscle pump, and changes in respiratory pressure.

Blood

Blood serves many diverse purposes in regulating normal body function. The three functions of primary importance to exercise and sport are

- transportation,
- temperature regulation, and
- acid–base (pH) balance.

We are most familiar with blood's transporting functions, delivering oxygen and fuel substrates and carrying away metabolic by-products. In addition, blood is critical in temperature regulation during physical activity because it picks up heat from the exercising muscle and delivers it to the skin, where it can be dissipated to the environment (see [chapter 12](#)). Blood also buffers the acids produced by anaerobic metabolism and maintains proper pH for metabolic processes (see [chapters 2](#) and [7](#)).

Blood Volume and Composition

The total volume of blood in the body varies considerably with an individual's size, body composition, and state of training. Larger blood volumes are associated with greater lean body mass and higher levels of endurance training. The blood volume of people of average body size and normal physical activity generally ranges from 5 to 6 L in men and 4 to 5 L in women.

Blood is composed of plasma and formed elements (see [figure 6.16](#)). Plasma normally constitutes about 55% to 60% of total blood volume but can decrease by 10% of its normal amount or more with intense exercise in hot conditions, or can increase by 10% or more with endurance training or acclimation to heat. Approximately 90% of the plasma volume is water; 7% consists of plasma proteins; and the remaining 3% includes cellular nutrients, electrolytes, enzymes, hormones, antibodies, and wastes.

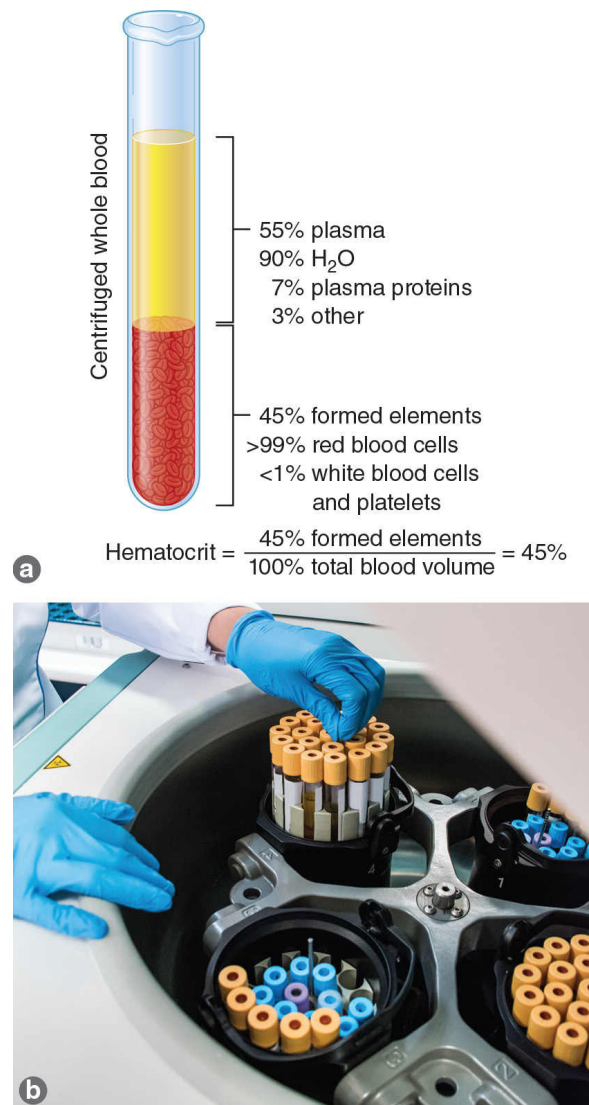


FIGURE 6.16 (a) The composition of whole blood, illustrating the plasma volume (fluid portion) and the cellular volume (red cells, white cells, and platelets) after the blood sample has been centrifuged to separate its components. (b) A centrifuge.

The formed elements, which normally constitute the other 40% to 45% of total blood volume, are the red blood cells (erythrocytes), white blood cells (leukocytes), and platelets (thrombocytes). Red blood cells constitute more than 99% of the formed-element volume; white blood cells and platelets together account for less than 1%. The percentage of the total blood volume composed of cells or formed elements is referred to as the **hematocrit**. Hematocrit varies among individuals, but a normal range is 41% to 50% in adult men and 36% to 44% in adult women.

White blood cells protect the body from infection either by directly destroying the invading agents through phagocytosis (ingestion) or by forming antibodies to destroy them. Adults have about 7,000 white blood cells per cubic millimeter of blood.

The remaining formed elements are the blood platelets. These are cell fragments that are required for blood coagulation (clotting), which prevents excessive blood loss. Exercise physiologists are most concerned with red blood cells.

Red Blood Cells

Mature red blood cells (erythrocytes) have no nucleus, so they cannot reproduce as other cells can. They must be replaced with new cells on a recurring basis, a process called **hematopoiesis**. The normal life span of a red blood cell is about 4 months. Thus, these cells are continuously produced and destroyed at equal rates. This balance is very important, because adequate oxygen delivery to tissues depends on having a sufficient number of red blood cells to transport oxygen. Decreases in their number or function can hinder oxygen delivery and thus affect exercise performance.

When we donate blood, the removal of one unit, or nearly 500 ml, represents approximately an 8% to 10% reduction in both the total blood volume and the number of circulating red blood cells. Donors are advised to drink plenty of fluids. Because plasma is primarily water, simple fluid replacement returns plasma volume to normal within 24 to 48 h. However, it takes at least 6 weeks to reconstitute the red blood cells because they must go through full development before they are functional. Blood loss greatly compromises the performance of endurance athletes by reducing oxygen delivery capacity.

Red blood cells transport oxygen, which is primarily bound to hemoglobin. **Hemoglobin** is composed of a protein (globin) and a pigment (heme). Heme contains iron, which binds oxygen. Each red blood cell contains approximately 250 million hemoglobin molecules, each able to bind four oxygen molecules—so each red blood cell can bind up to a billion molecules of oxygen! There is an average of 15 g of hemoglobin per 100 ml of whole blood. Each gram of hemoglobin can combine with 1.33 ml of oxygen, so as much as 20 ml of oxygen

can be bound for each 100 ml of blood. Therefore, when arterial blood is saturated with oxygen, it has an oxygen-carrying capacity of 20 ml of oxygen per 100 ml of blood.

Blood Viscosity

Viscosity refers to the thickness of the blood. Recall from our discussion of vascular resistance that the more viscous a fluid, the more resistant it is to flow. Syrup is more viscous than water and thus flows more slowly when poured. The viscosity of blood is normally about twice that of water and increases as hematocrit increases.

Because of oxygen transport by the red blood cells, an increase in their number would be expected to maximize oxygen transport. But if an increase in red blood cell count is not accompanied by a similar increase in plasma volume, blood viscosity and vascular resistance will increase, which could result in reduced blood flow. This generally is not a problem unless the hematocrit reaches 60% or more.

Conversely, the combination of a low hematocrit with a high plasma volume, which decreases the blood's viscosity, appears to have certain benefits for the blood's transport function because the blood can flow more easily. Unfortunately, a low hematocrit frequently results from a reduced red blood cell count, as in diseases such as anemia. Under these circumstances, the blood can flow easily, but it contains fewer carriers, so oxygen transport is impeded. For optimal physical performance, a low-normal hematocrit with a normal or slightly elevated number of red blood cells is desirable. This combination facilitates oxygen transport. Many endurance athletes achieve this combination as part of their cardiovascular system's normal adaptation to training. This adaptation is discussed in [chapter 11](#).

In Review

- Blood is about 55% to 60% plasma and 40% to 45% formed elements. Red blood cells compose about 99% of the formed elements.
- The hematocrit is the ratio of the formed elements in the blood (red cells, white cells, and platelets) to the total blood volume. An average hematocrit for adult men is 42% and for adult women is 38%.

- Oxygen is transported primarily by binding to the hemoglobin in red blood cells.
 - During endurance training, athletes respond with both a higher red cell volume (RCV) and an expanded plasma volume (PV). Since the PV increase is higher than the RCV increase, the hematocrit in these athletes tends to be somewhat lower than that of sedentary individuals.
 - As blood viscosity increases, so does resistance to flow. Increasing the number of red blood cells is advantageous to aerobic performance but only up to the point (a hematocrit approaching 60%) where viscosity limits flow.
-

IN CLOSING

In this chapter, we reviewed the structure and function of the cardiovascular system. We learned how blood flow and blood pressure are regulated to meet the body's needs and explored the role of the cardiovascular system in transporting and delivering oxygen and nutrients to the body's cells while clearing away metabolic wastes, including carbon dioxide. Knowing how substances are moved within the body, we now look more closely at the transport of oxygen and carbon dioxide. In the next chapter, we explore the role of the respiratory system in delivering oxygen to, and removing carbon dioxide from, the cells of the body.

KEY TERMS

arteries
arterioles
atherosclerosis
atrioventricular (AV) node
baroreceptor
bradycardia
capillaries
cardiac cycle
cardiac output ($\dot{Q}$)
chemoreceptor
diastolic blood pressure (DBP)
ejection fraction (EF)
electrocardiogram (ECG)
electrocardiograph
end-diastolic volume (EDV)
end-systolic volume (ESV)
extrinsic neural control
functional sympatholysis
heart murmur
hematocrit
hematopoiesis
hemoglobin
intercalated disks
mean arterial pressure (MAP)
mechanoreceptors
muscle pump

myocardium
pericardium
premature ventricular contraction (PVC)
Purkinje fibers
sinoatrial (SA) node
stroke volume (SV)
systolic blood pressure (SBP)
tachycardia
vasoconstriction
vasodilation
veins
ventricular fibrillation
ventricular tachycardia
venules

STUDY QUESTIONS

1. Describe the structure of the heart, the pattern of blood flow through the valves and chambers of the heart, how the heart as a muscle is supplied with blood, and what happens when the resting heart must suddenly supply an exercising body.
2. What events take place that allow the heart to contract, and how is heart rate controlled?
3. What is torsional contraction of the heart, and why is it important during exercise?
4. What is the difference between systole and diastole, and how do they relate to SBP and DBP?
5. What is the relationship between pressure, flow, and resistance?
6. How is blood flow to the various regions of the body controlled?
7. How does muscle blood flow increase during exercise despite increased sympathetic nerve activity that favors vasoconstriction?
8. Describe the three important mechanisms for returning blood back to the heart when someone is exercising in an upright position.
9. Describe the primary functions of blood.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

